ABCD ADHD menarche analysis: Output for longitudinal models

2025-07-22

Model 1 longitudinal output

```
    Estimate   Est.Error      Q2.5     Q97.5
R2 0.7416968 0.006957391 0.7274837 0.7548127


 Family: poisson
  Links: mu = log
Formula: cbcl_scr_syn_attention_r_wave_3 ~ menarche_status_p_wave_2 + age_years_c_wave_2 + et
   Data: imp_df_wide (Number of observations: 4740)
  Draws: 2 chains, each with iter = 5000; warmup = 2500; thin = 1;
         total post-warmup draws = 5000

Multilevel Hyperparameters:
~family_id (Number of levels: 4111)
              Estimate Est.Error l-95% CI u-95% CI   Rhat Bulk_ESS Tail_ESS
sd(Intercept)   0.8511    0.0194   0.8137   0.8896 1.0009     1858     2966

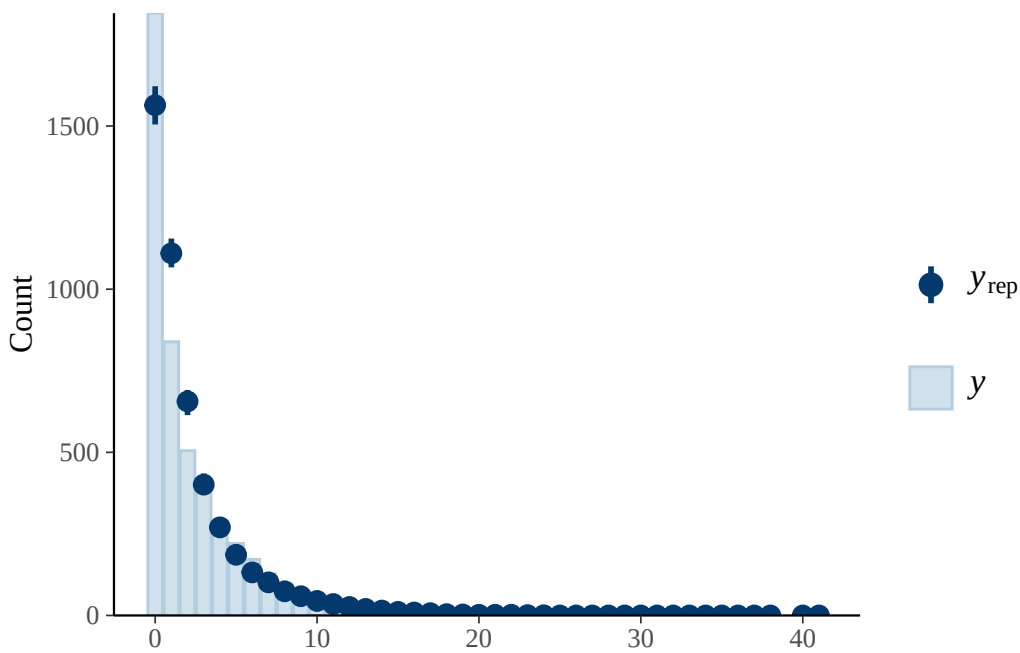
~site_id (Number of levels: 22)
              Estimate Est.Error l-95% CI u-95% CI   Rhat Bulk_ESS Tail_ESS
sd(Intercept)   0.0959    0.0345   0.0312   0.1695 1.0030      903      822

Regression Coefficients:
                                 Estimate Est.Error l-95% CI u-95% CI   Rhat
Intercept                          0.2566    0.0417   0.1730   0.3384 1.0005
menarche_status_p_wave_2Y          0.0987    0.0386   0.0243   0.1746 1.0007
age_years_c_wave_2                -0.0570    0.0284  -0.1129  -0.0024 1.0000
ethnicity2                        -0.0515    0.0598  -0.1706   0.0683 1.0002
ethnicity3                        -0.0316    0.0555  -0.1414   0.0814 1.0003
```

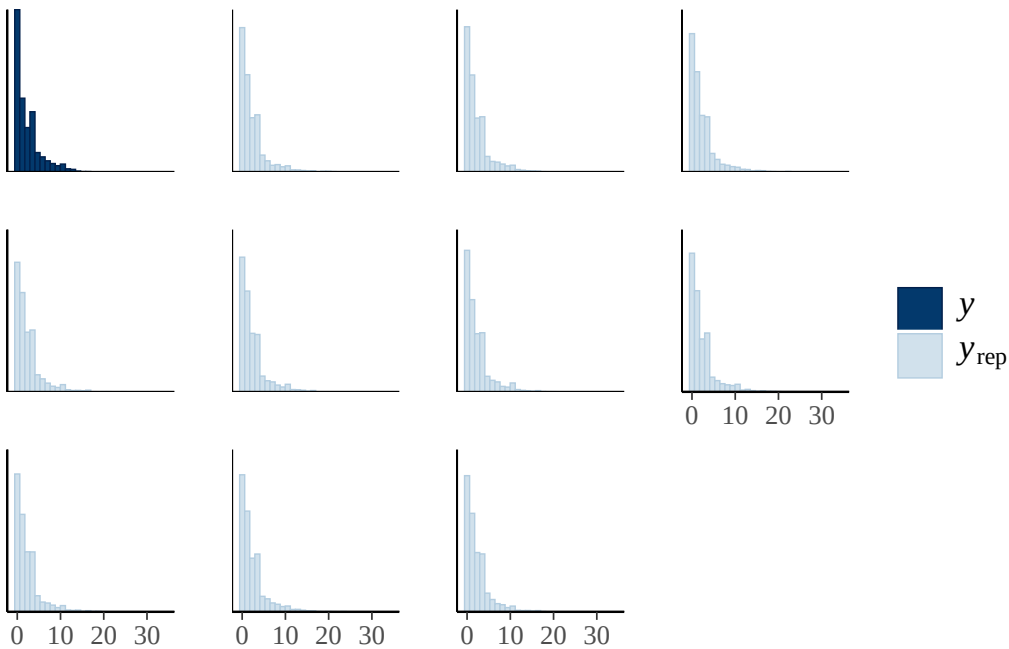
```
ethnicity4                          -0.2362    0.1322  -0.4914   0.0262 1.0003
ethnicity5                           0.0858    0.0606  -0.0319   0.2060 1.0001
inr_c_wave_0                         0.0128    0.0062   0.0011   0.0250 1.0008
cbcl_scr_syn_attention_r_c_wave_0    0.2127    0.0051   0.2029   0.2227 1.0012
                                   Bulk_ESS Tail_ESS
Intercept                              2400     2961
menarche_status_p_wave_2Y              2414     3219
age_years_c_wave_2                     2361     2808
ethnicity2                             2129     3365
ethnicity3                             2463     3231
ethnicity4                             3137     3730
ethnicity5                             2138     2952
inr_c_wave_0                           2478     3590
cbcl_scr_syn_attention_r_c_wave_0      1995     2841

Draws were sampled using sampling(NUTS). For each parameter, Bulk_ESS
and Tail_ESS are effective sample size measures, and Rhat is the potential
scale reduction factor on split chains (at convergence, Rhat = 1).
```

Model 1 long posterior predictive diagnostics




```
`stat_bin()` using `bins = 30`. Pick better value with `binwidth`.
```

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`stat_bin()` using `bins = 30`. Pick better value with `binwidth`.
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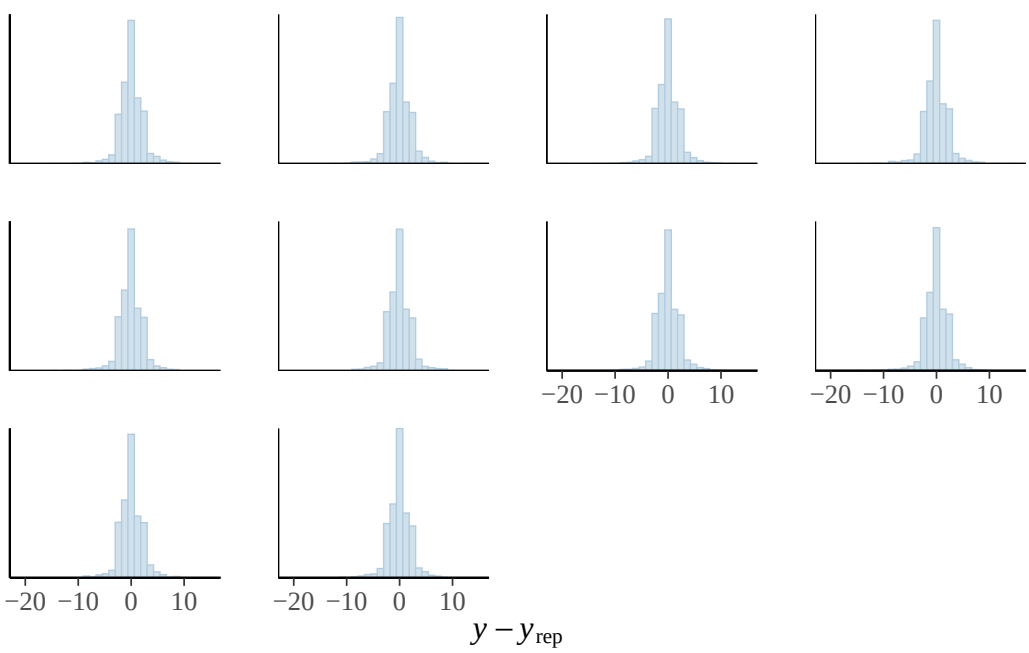
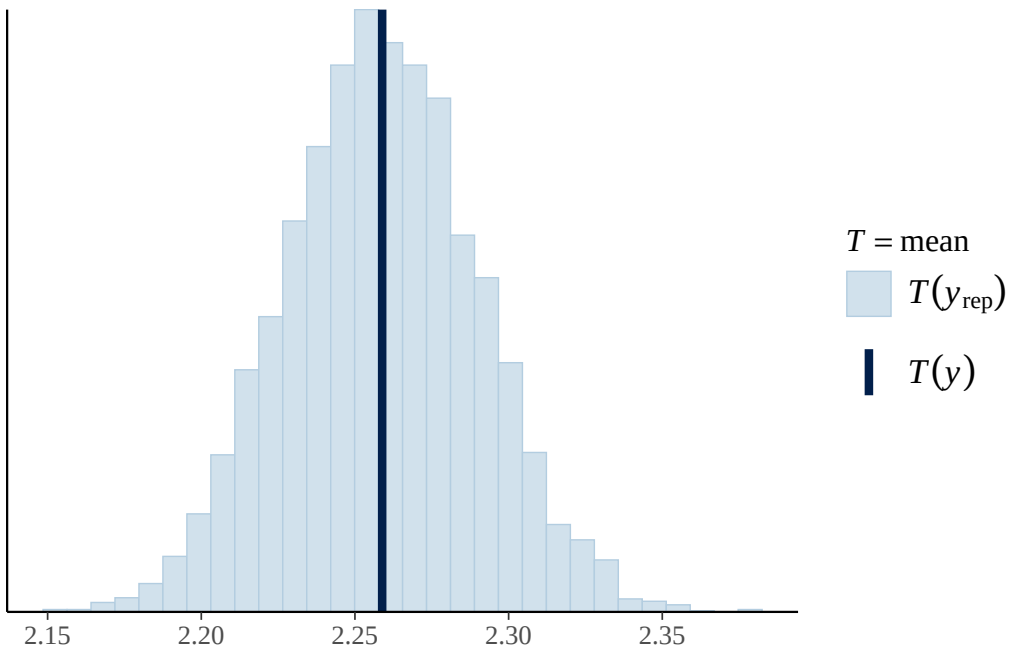
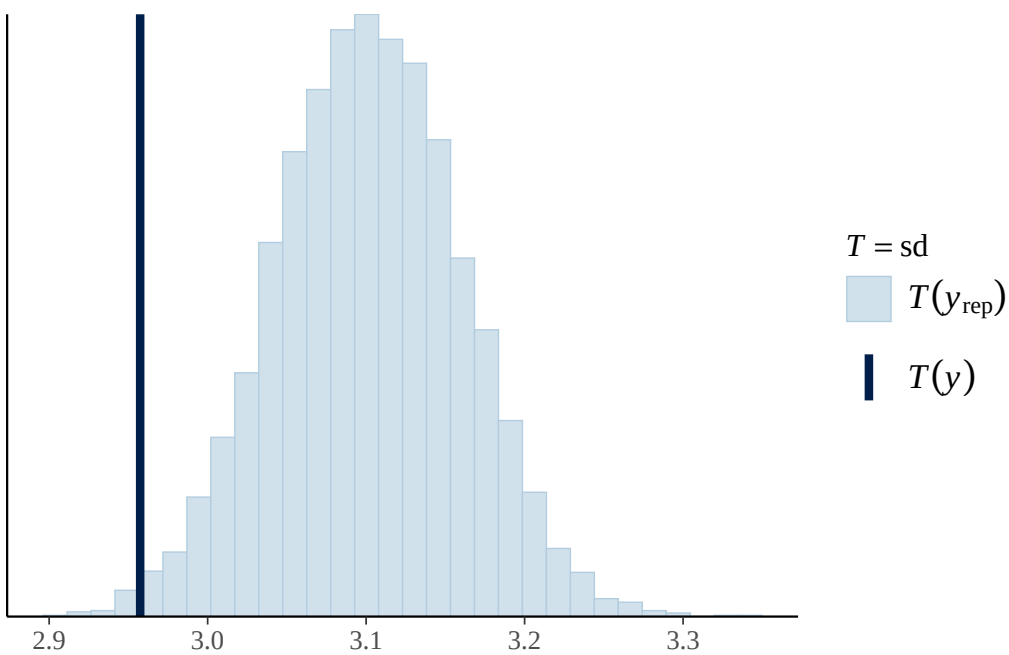
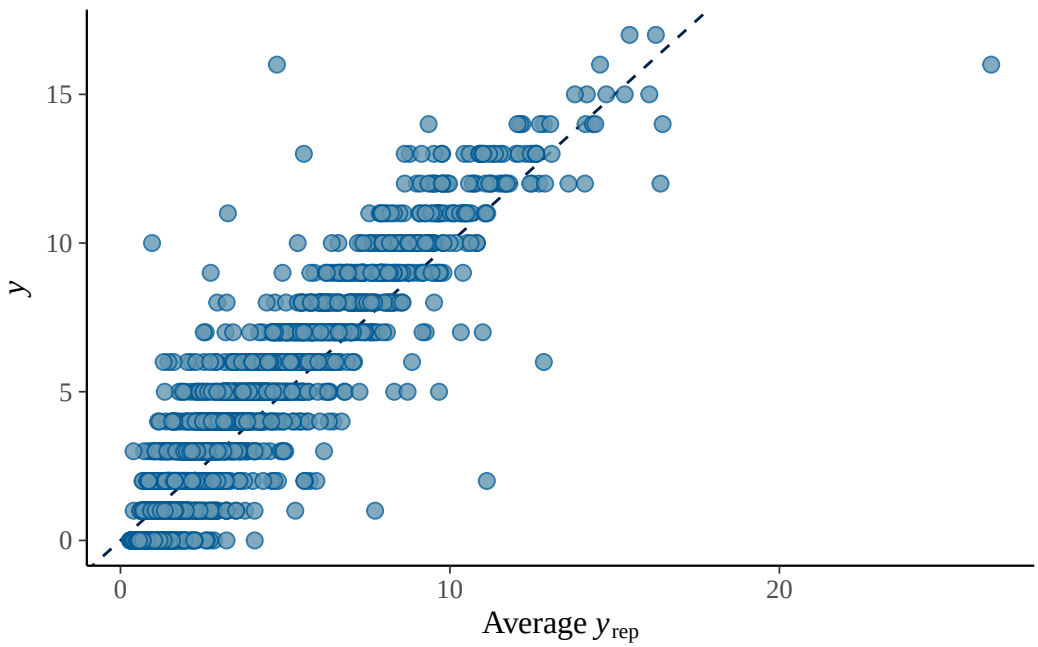



```
`stat_bin()` using `bins = 30`. Pick better value with `binwidth`.
```




```
`stat_bin()` using `bins = 30`. Pick better value with `binwidth`.
```





Model 1b longitudinal output

```
    Estimate  Est.Error      Q2.5     Q97.5
R2 0.7423426 0.00691692 0.7281641 0.7556766


 Family: poisson
  Links: mu = log
Formula: cbcl_scr_syn_attention_r_wave_3 ~ pds_f4_p_c_wave_2 + age_years_c_wave_2 + ethnicity
   Data: imp_df_wide (Number of observations: 4820)
  Draws: 2 chains, each with iter = 5000; warmup = 2500; thin = 1;
         total post-warmup draws = 5000

Multilevel Hyperparameters:
~family_id (Number of levels: 4177)
              Estimate Est.Error l-95% CI u-95% CI   Rhat Bulk_ESS Tail_ESS
sd(Intercept)   0.8514    0.0193   0.8148   0.8906 1.0004     1495     2604

~site_id (Number of levels: 22)
              Estimate Est.Error l-95% CI u-95% CI   Rhat Bulk_ESS Tail_ESS
sd(Intercept)   0.1001    0.0334   0.0381   0.1706 1.0031      821      775

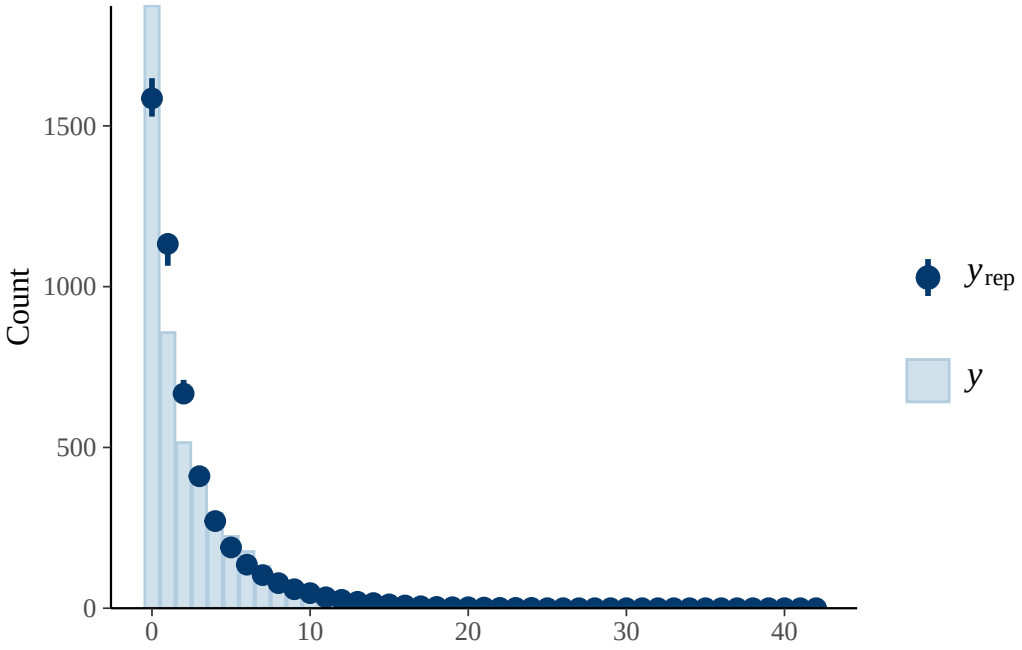
Regression Coefficients:
```

	Estimate	Est.Error	l-95% CI	u-95% CI	Rhat
Intercept	0.2732	0.0398	0.1950	0.3504	1.0009
pds_f4_p_c_wave_2	0.1088	0.0280	0.0543	0.1650	1.0000
age_years_c_wave_2	-0.0535	0.0272	-0.1071	0.0003	1.0004
ethnicity2	-0.0609	0.0600	-0.1789	0.0549	1.0009
ethnicity3	-0.0334	0.0556	-0.1406	0.0782	1.0003
ethnicity4	-0.2563	0.1329	-0.5162	-0.0003	1.0004
ethnicity5	0.0643	0.0579	-0.0472	0.1810	1.0003
inr_c_wave_0	0.0117	0.0061	-0.0003	0.0232	1.0035
cbcl_scr_syn_attention_r_c_wave_0	0.2112	0.0050	0.2013	0.2209	1.0006

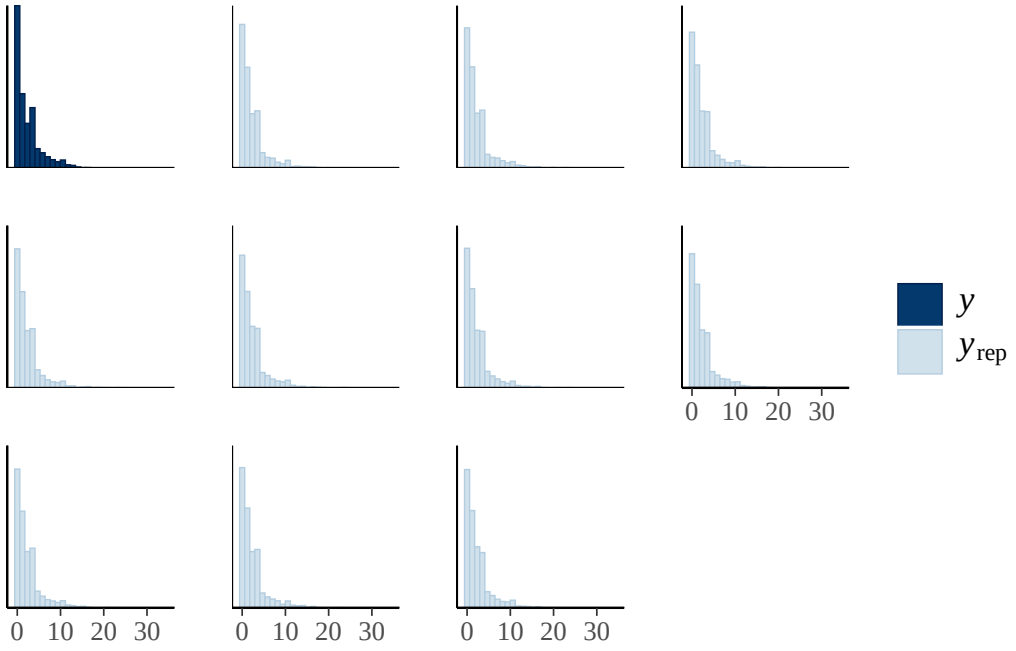
	Bulk_ESS	Tail_ESS
Intercept	2266	2762
pds_f4_p_c_wave_2	2723	3513
age_years_c_wave_2	2350	3036
ethnicity2	2030	3079
ethnicity3	1743	2662
ethnicity4	2638	3080
ethnicity5	2211	3288
inr_c_wave_0	1964	2680
cbcl_scr_syn_attention_r_c_wave_0	1508	2468

Draws were sampled using sampling(NUTS). For each parameter, Bulk_ESS and Tail_ESS are effective sample size measures, and Rhat is the potential scale reduction factor on split chains (at convergence, Rhat = 1).

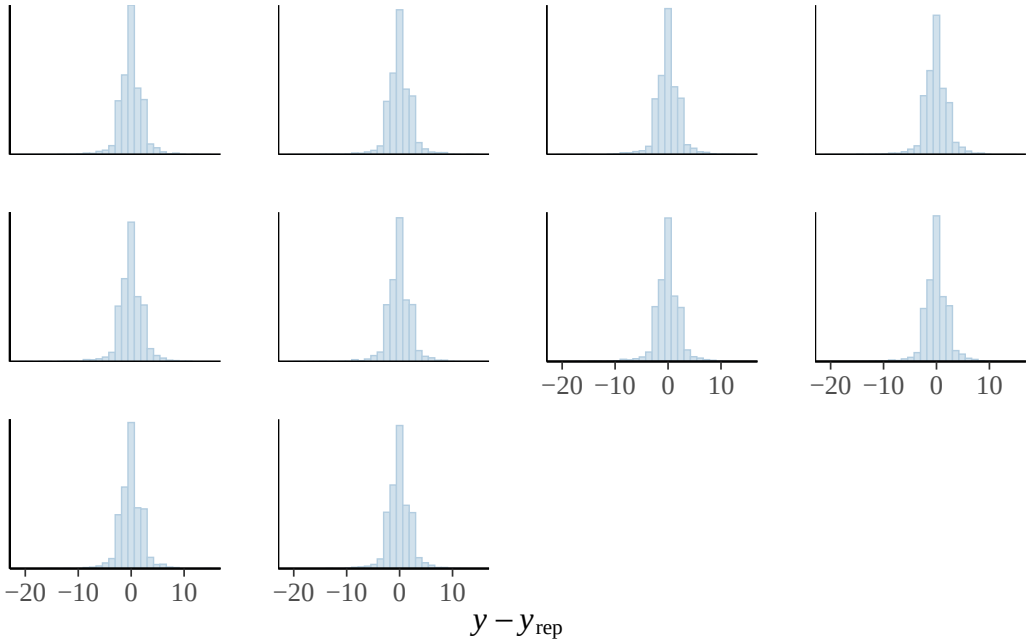
Model 1b long posterior predictive diagnostics



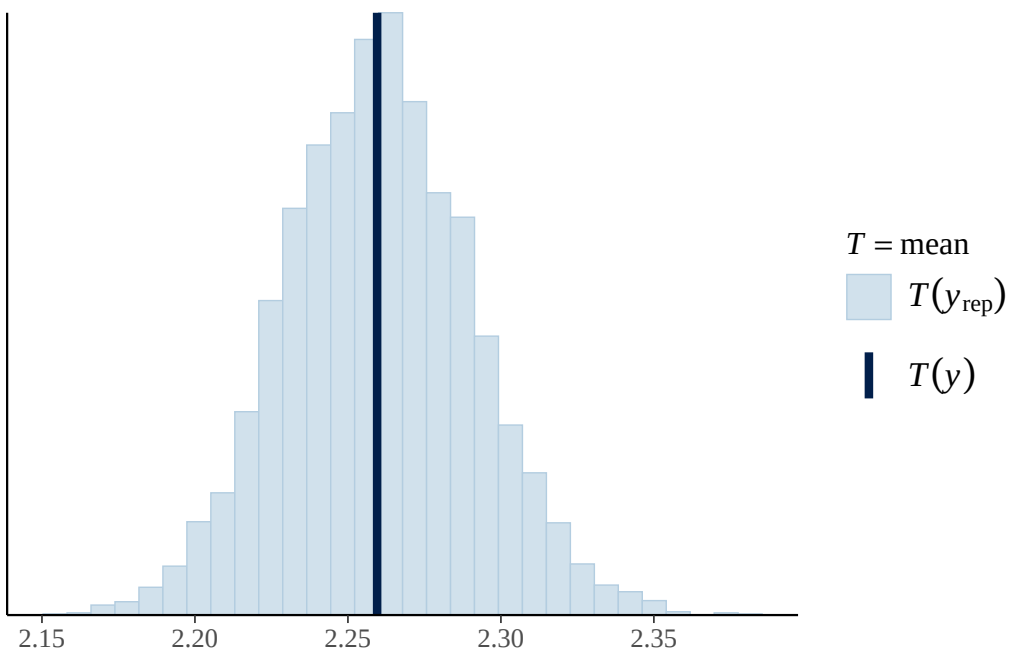

```
`stat_bin()` using `bins = 30`. Pick better value with `binwidth`.
```



```
`stat_bin()` using `bins = 30`. Pick better value with `binwidth`.
```




```
`stat_bin()` using `bins = 30`. Pick better value with `binwidth`.
```



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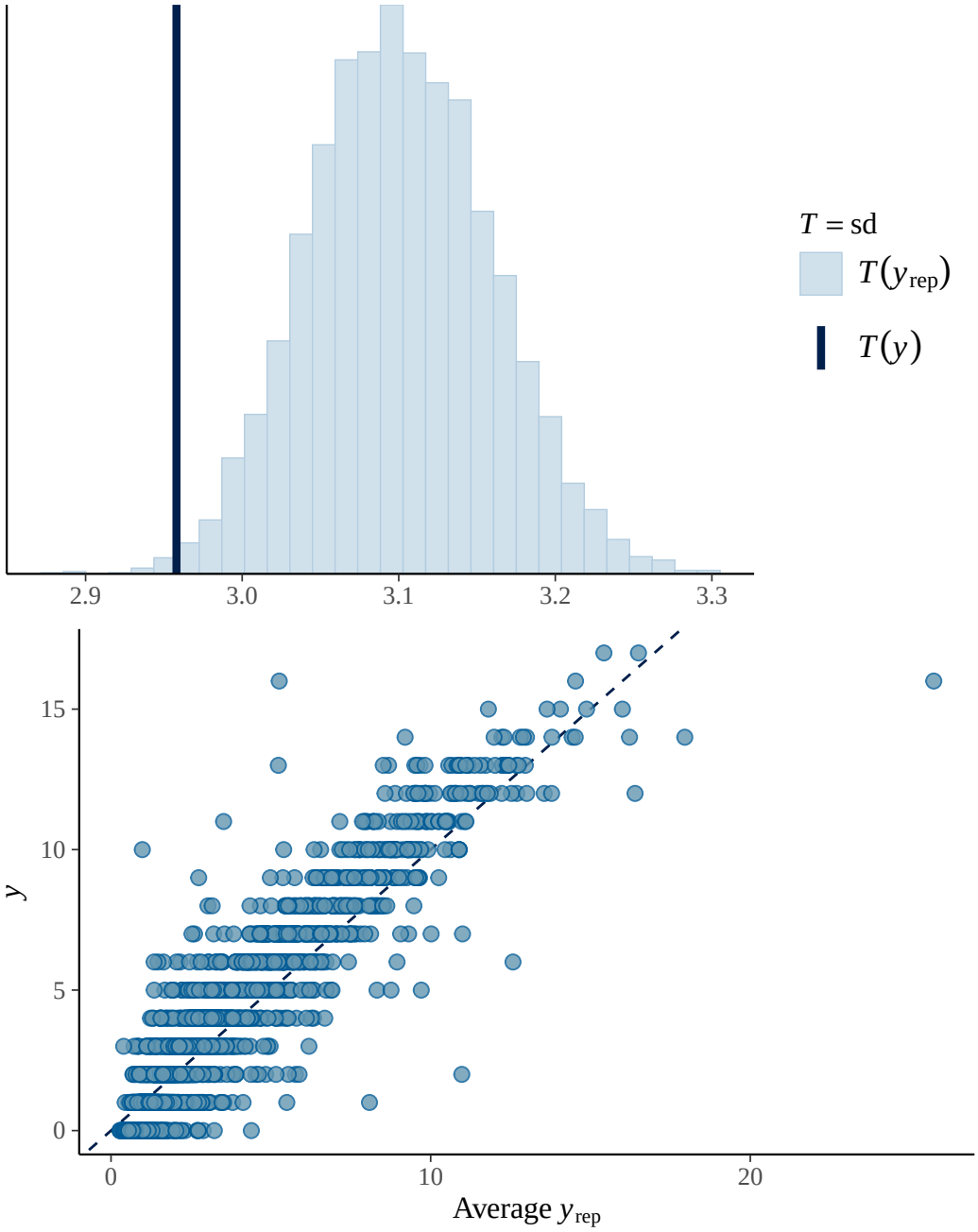
`stat_bin()` using `bins = 30`. Pick better value with `binwidth`.
```



Model 2 longitudinal output

```
    Estimate   Est.Error      Q2.5     Q97.5
R2 0.8351107 0.004311799 0.8265066 0.8433876


 Family: poisson
  Links: mu = log
Formula: cbcl_scr_syn_internal_r_wave_3 ~ menarche_status_p_wave_2 * adhd_diagnosis_wave_0 +
   Data: imp_df_wide (Number of observations: 4740)
  Draws: 2 chains, each with iter = 5000; warmup = 2500; thin = 1;
         total post-warmup draws = 5000

Multilevel Hyperparameters:
~family_id (Number of levels: 4111)
              Estimate Est.Error l-95% CI u-95% CI   Rhat Bulk_ESS Tail_ESS
sd(Intercept)   0.8294    0.0142   0.8023   0.8575 1.0020     1021     2247

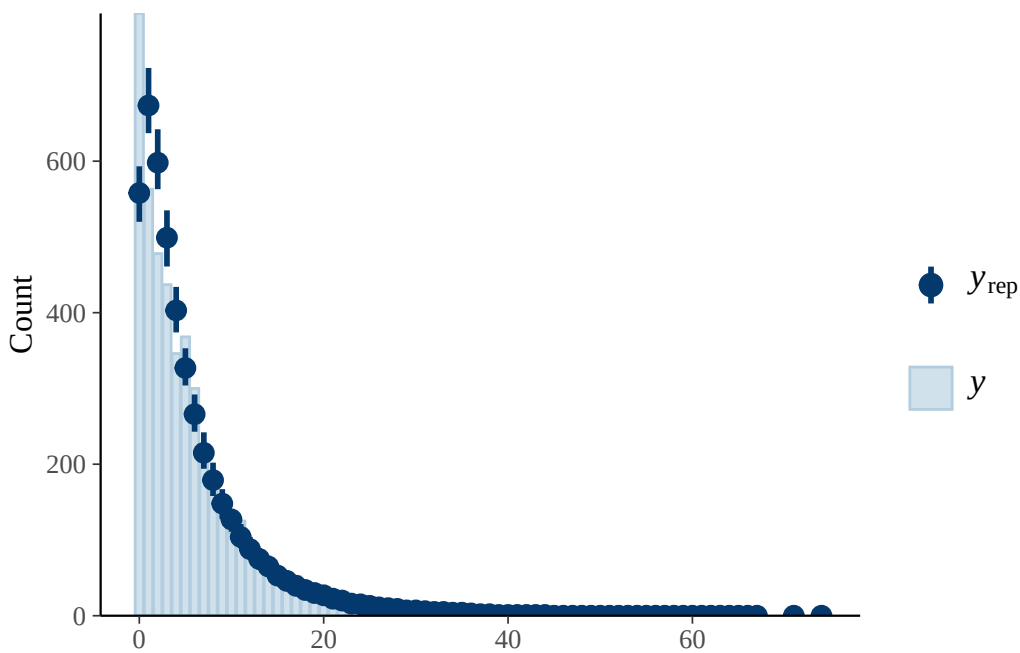
~site_id (Number of levels: 22)
              Estimate Est.Error l-95% CI u-95% CI   Rhat Bulk_ESS Tail_ESS
sd(Intercept)   0.1097    0.0318   0.0547   0.1789 1.0065      559      954

Regression Coefficients:
                                                Estimate Est.Error l-95% CI
Intercept                                         1.2491    0.0376   1.1774
menarche_status_p_wave_2Y                         0.1797    0.0332   0.1137
adhd_diagnosis_wave_0                             0.2526    0.0400   0.1746
age_years_c_wave_2                                0.0043    0.0221  -0.0395
ethnicity2                                       -0.4049    0.0531  -0.5101
ethnicity3                                       -0.0763    0.0474  -0.1681
ethnicity4                                       -0.3716    0.1069  -0.5857
ethnicity5                                        0.0989    0.0500  -0.0010
inr_c_wave_0                                      0.0148    0.0051   0.0047
cbcl_scr_syn_internal_r_c_wave_0                  0.0763    0.0024   0.0715
menarche_status_p_wave_2Y:adhd_diagnosis_wave_0   0.0611    0.0609  -0.0563
                                                u-95% CI   Rhat Bulk_ESS
Intercept                                         1.3235 1.0026     1081
menarche_status_p_wave_2Y                         0.2440 1.0010     1204
adhd_diagnosis_wave_0                             0.3314 1.0006     1142
age_years_c_wave_2                                0.0466 1.0006     1228
ethnicity2                                       -0.3035 1.0004     1189
ethnicity3                                        0.0158 1.0046      792
ethnicity4                                       -0.1638 1.0016     1466
```

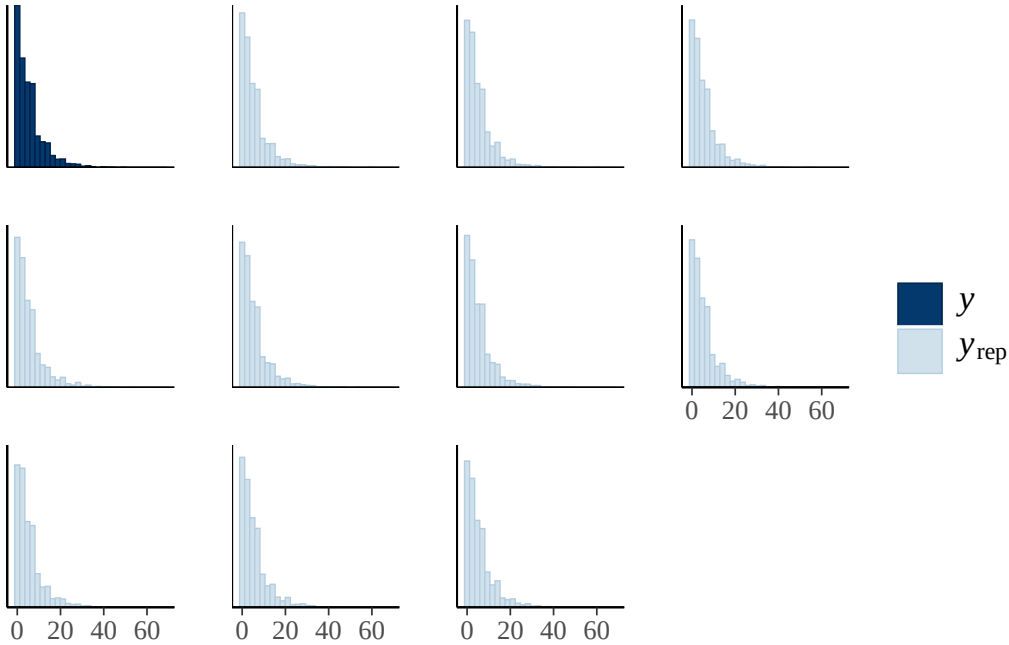
```
ethnicity5                                          0.1966 1.0004      894
inr_c_wave_0                                        0.0246 1.0005     1127
cbcl_scr_syn_internal_r_c_wave_0                    0.0808 1.0023      980
menarche_status_p_wave_2Y:adhd_diagnosis_wave_0     0.1812 1.0032     1058
                                                  Tail_ESS
Intercept                                            2236
menarche_status_p_wave_2Y                            2399
adhd_diagnosis_wave_0                                1829
age_years_c_wave_2                                   2052
ethnicity2                                           2534
ethnicity3                                           1691
ethnicity4                                           1734
ethnicity5                                           1666
inr_c_wave_0                                         1824
cbcl_scr_syn_internal_r_c_wave_0                     1749
menarche_status_p_wave_2Y:adhd_diagnosis_wave_0      1756

Draws were sampled using sampling(NUTS). For each parameter, Bulk_ESS
and Tail_ESS are effective sample size measures, and Rhat is the potential
scale reduction factor on split chains (at convergence, Rhat = 1).
```

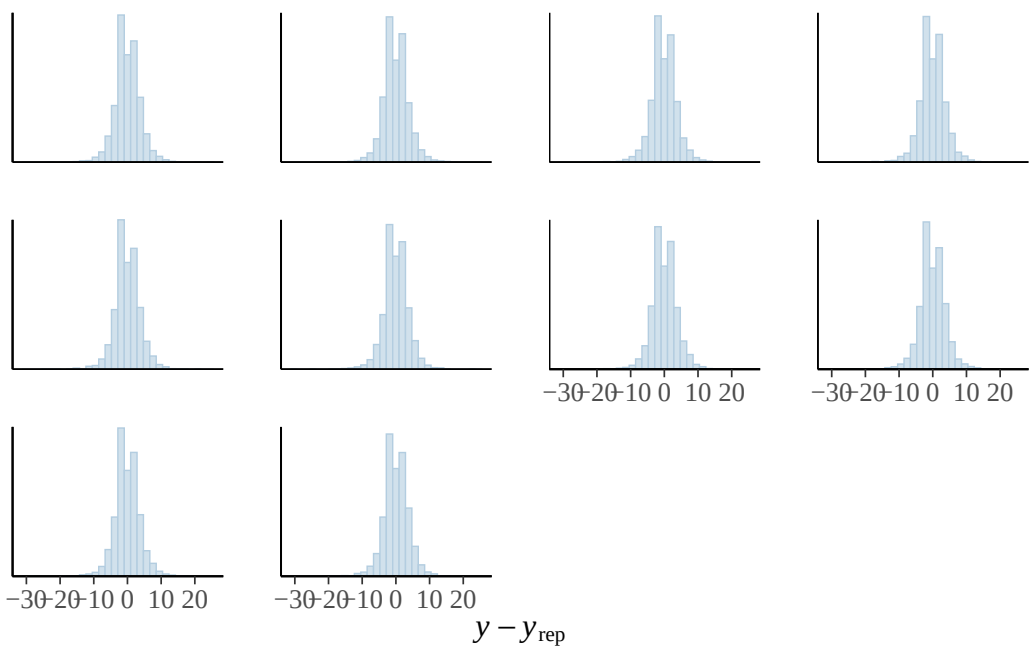
Model 2 long posterior predictive diagnostics



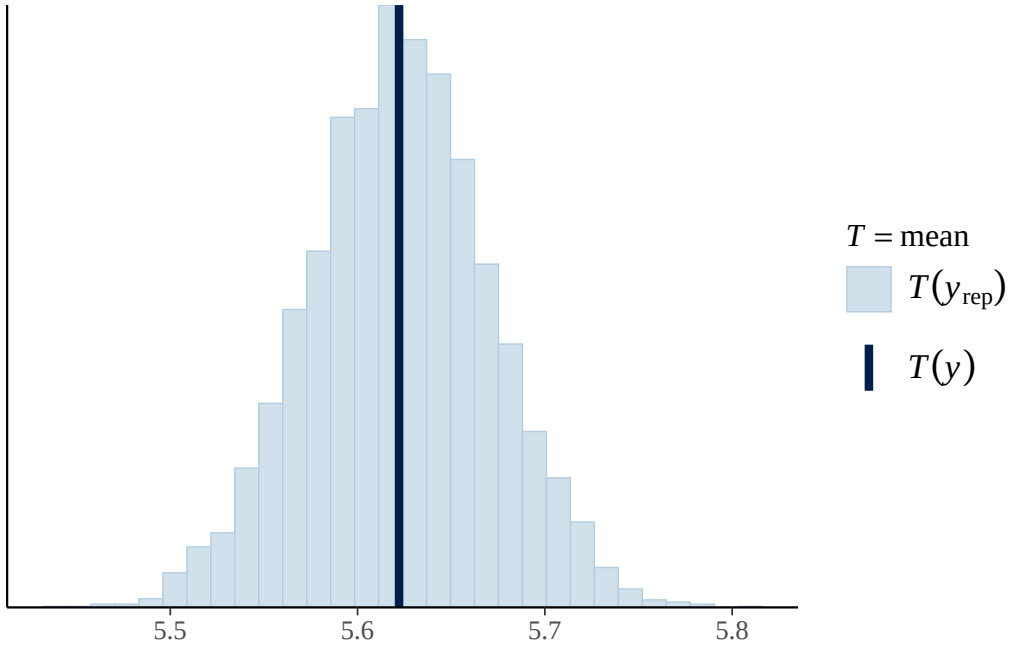
```
`stat_bin()` using `bins = 30`. Pick better value with `binwidth`.
```



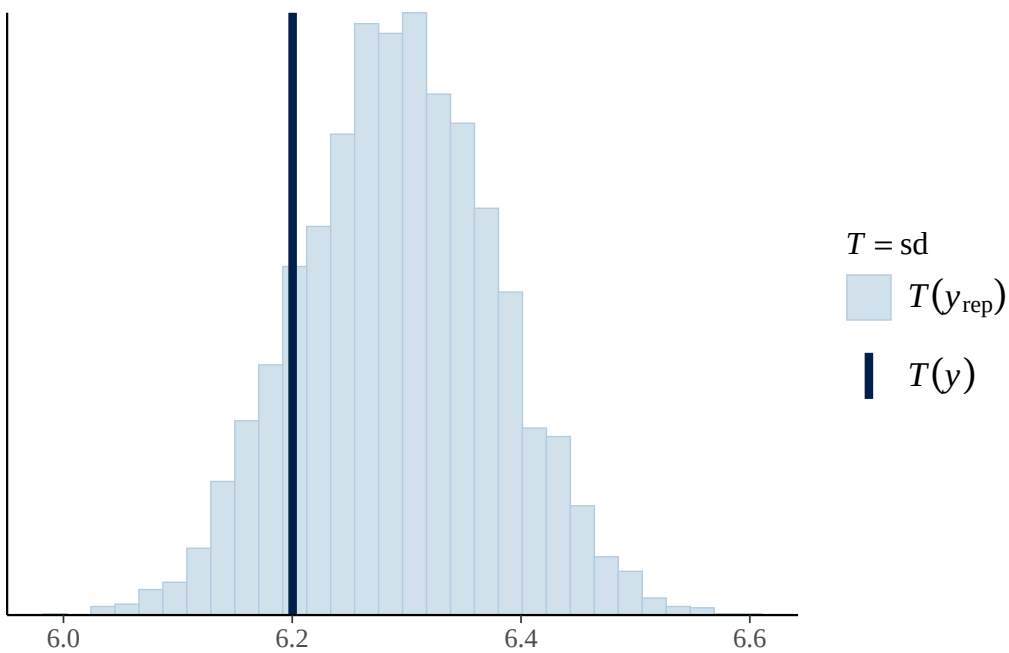

```
`stat_bin()` using `bins = 30`. Pick better value with `binwidth`.
```

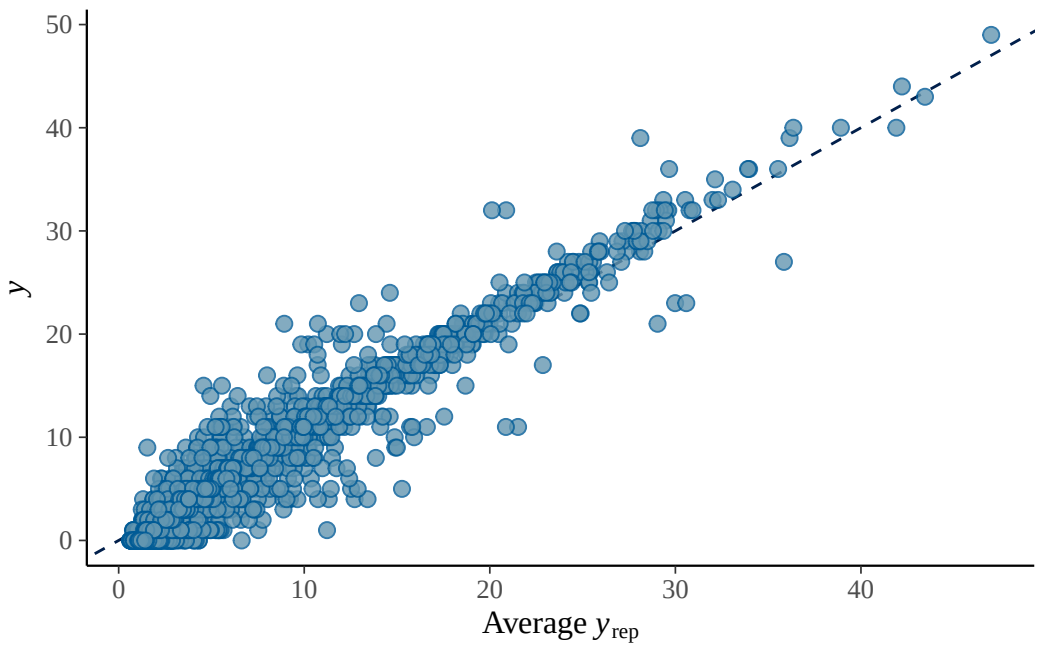


```
`stat_bin()` using `bins = 30`. Pick better value with `binwidth`.
```




```
`stat_bin()` using `bins = 30`. Pick better value with `binwidth`.
```





Model 2b longitudinal output

```
   Estimate   Est.Error     Q2.5     Q97.5
R2 0.835986 0.004173998 0.827794 0.8440491


 Family: poisson
  Links: mu = log
Formula: cbcl_scr_syn_internal_r_wave_3 ~ pds_f4_p_c_wave_2 * adhd_diagnosis + age_years_c_wa
   Data: imp_df_wide (Number of observations: 4820)
  Draws: 2 chains, each with iter = 5000; warmup = 2500; thin = 1;
         total post-warmup draws = 5000

Multilevel Hyperparameters:
~family_id (Number of levels: 4177)
              Estimate Est.Error l-95% CI u-95% CI   Rhat Bulk_ESS Tail_ESS
sd(Intercept)   0.8348    0.0143   0.8074   0.8628 1.0006     1179     2134

~site_id (Number of levels: 22)
              Estimate Est.Error l-95% CI u-95% CI   Rhat Bulk_ESS Tail_ESS
sd(Intercept)   0.1082    0.0295   0.0578   0.1724 1.0075      859     1214

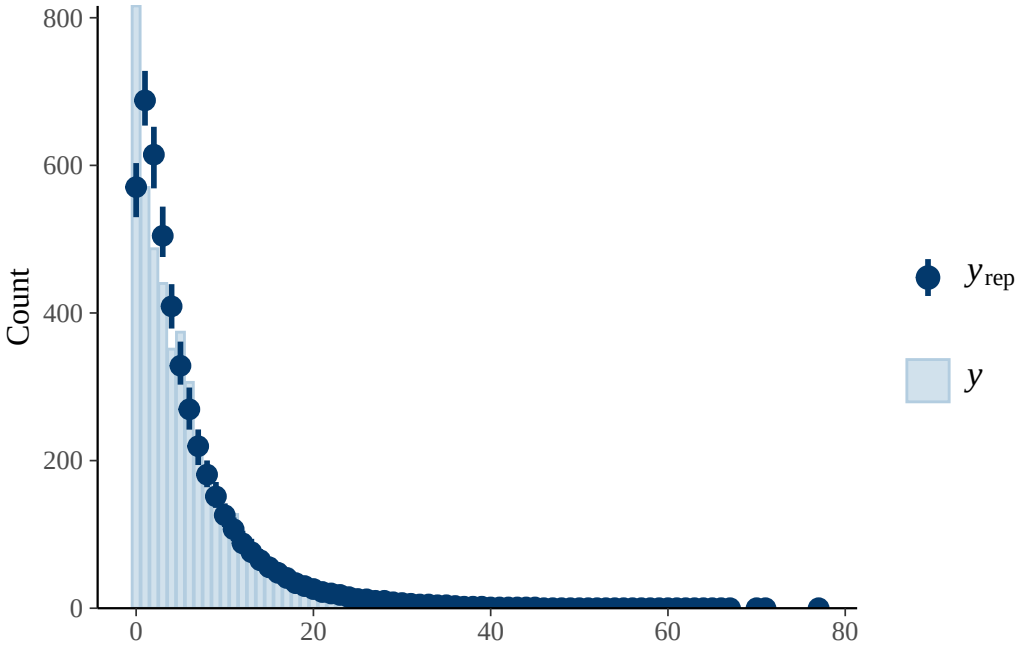
Regression Coefficients:
```

	Estimate	Est.Error	l-95% CI	u-95% CI	Rhat
Intercept	1.2777	0.0373	1.2017	1.3502	1.0004
pds_f4_p_c_wave_2	0.1166	0.0238	0.0702	0.1632	1.0026
adhd_diagnosis1	0.2419	0.0332	0.1775	0.3061	1.0002
age_years_c_wave_2	0.0277	0.0206	-0.0129	0.0687	1.0007
ethnicity2	-0.3832	0.0532	-0.4894	-0.2800	1.0008
ethnicity3	-0.0609	0.0463	-0.1509	0.0300	1.0023
ethnicity4	-0.3680	0.1081	-0.5791	-0.1584	1.0006
ethnicity5	0.0855	0.0507	-0.0146	0.1829	0.9999
inr_c_wave_0	0.0129	0.0052	0.0027	0.0233	1.0013
cbcl_scr_syn_internal_r_c_wave_0	0.0769	0.0025	0.0722	0.0819	1.0004
pds_f4_p_c_wave_2:adhd_diagnosis1	0.0381	0.0479	-0.0567	0.1297	1.0016

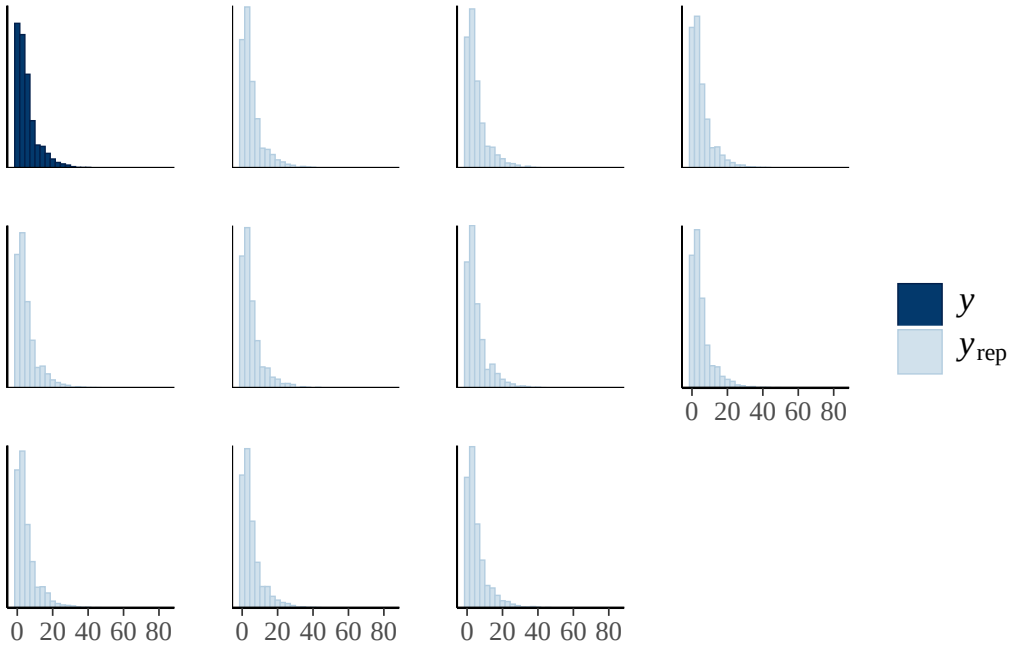
	Bulk_ESS	Tail_ESS
Intercept	1784	2428
pds_f4_p_c_wave_2	1970	2729
adhd_diagnosis1	1917	2958
age_years_c_wave_2	1900	2759
ethnicity2	1208	2040
ethnicity3	1208	2454
ethnicity4	1964	2978
ethnicity5	1202	1932
inr_c_wave_0	1339	2141
cbcl_scr_syn_internal_r_c_wave_0	1233	2248
pds_f4_p_c_wave_2:adhd_diagnosis1	1772	2298

Draws were sampled using sampling(NUTS). For each parameter, Bulk_ESS and Tail_ESS are effective sample size measures, and Rhat is the potential scale reduction factor on split chains (at convergence, Rhat = 1).

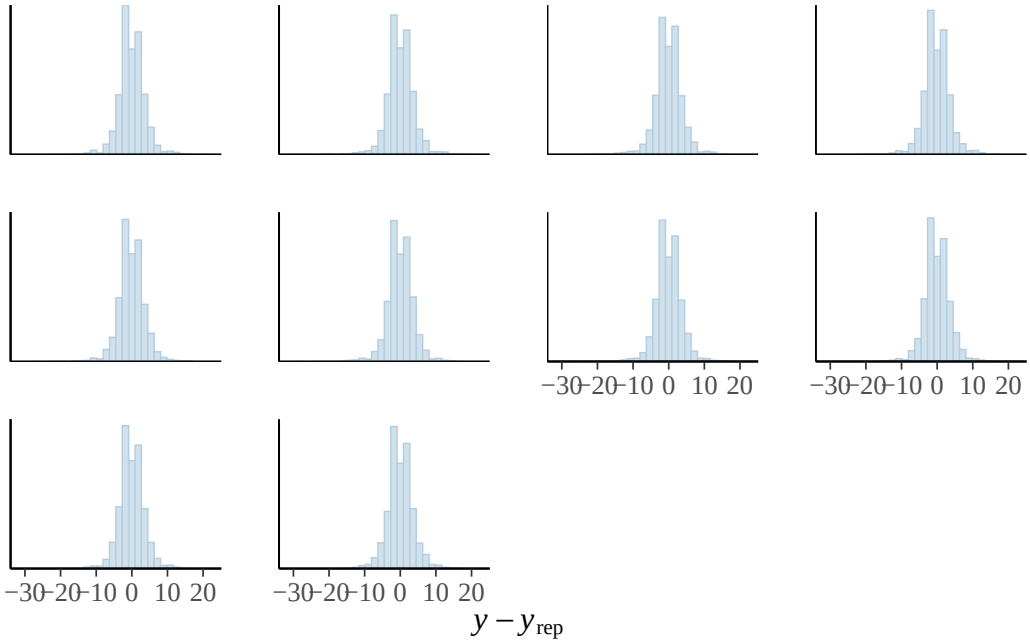
Model 2b long posterior predictive diagnostics



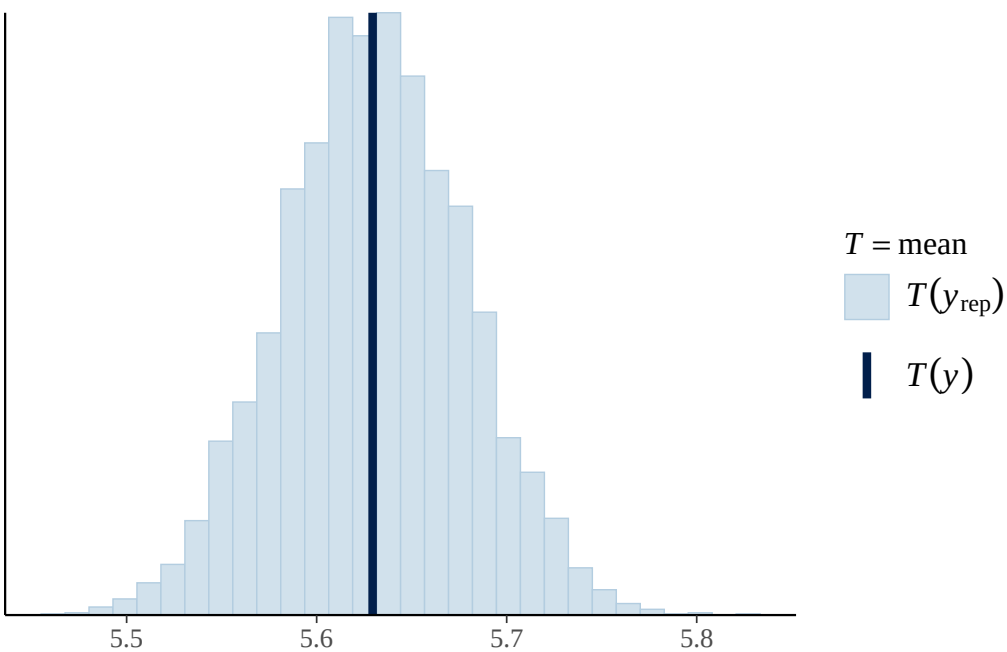

```
`stat_bin()` using `bins = 30`. Pick better value with `binwidth`.
```



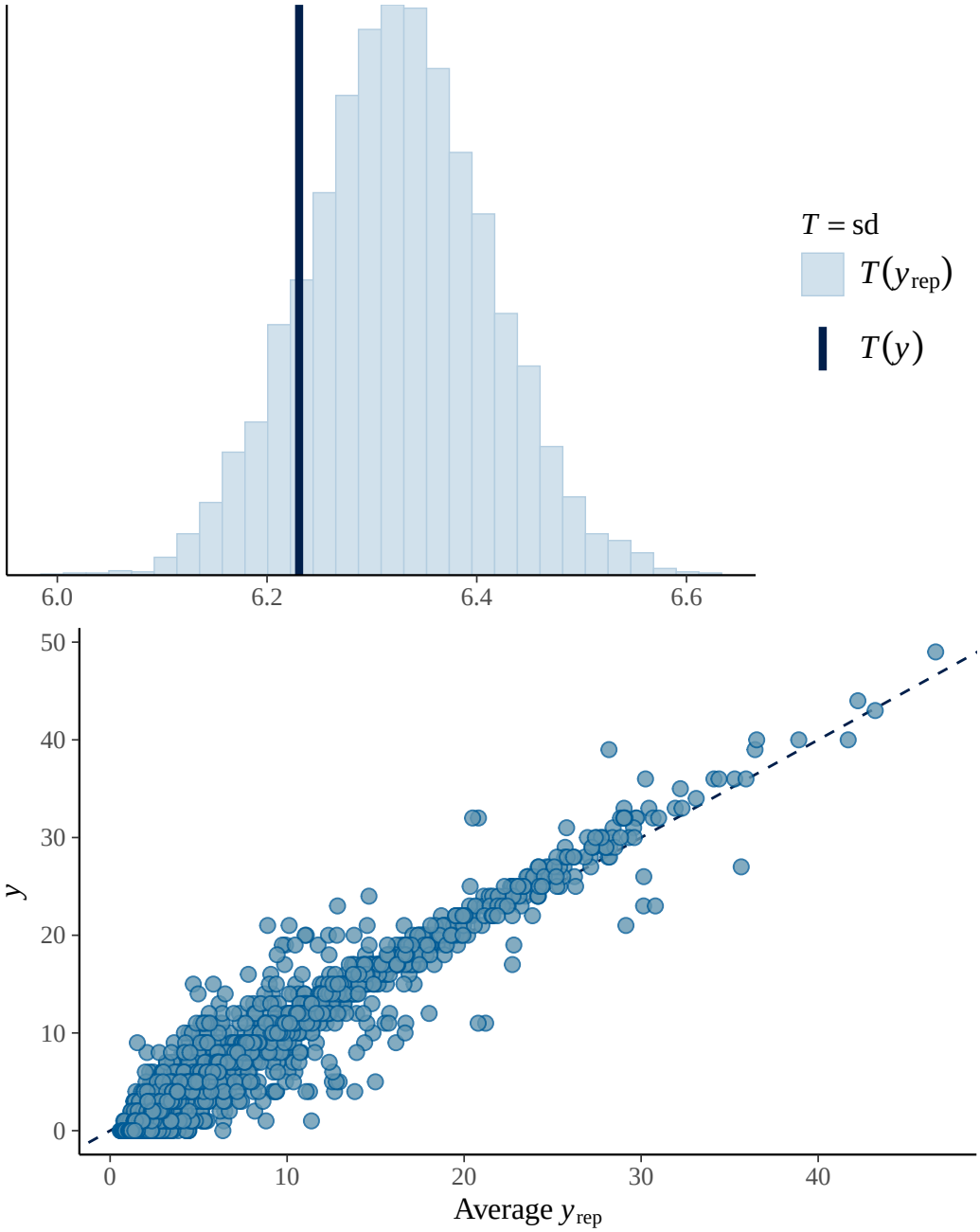
```
`stat_bin()` using `bins = 30`. Pick better value with `binwidth`.
```




```
`stat_bin()` using `bins = 30`. Pick better value with `binwidth`.
```



```
`stat_bin()` using `bins = 30`. Pick better value with `binwidth`.
```



Model 3 longitudinal output

```
    Estimate   Est.Error      Q2.5     Q97.5
R2 0.8260905 0.004777247 0.8164656 0.8350482


 Family: poisson
  Links: mu = log
Formula: cbcl_scr_syn_external_r_wave_3 ~ menarche_status_p_wave_2 * adhd_diagnosis + age_yea
   Data: imp_df_wide (Number of observations: 4740)
  Draws: 2 chains, each with iter = 5000; warmup = 2500; thin = 1;
         total post-warmup draws = 5000

Multilevel Hyperparameters:
~family_id (Number of levels: 4111)
              Estimate Est.Error l-95% CI u-95% CI   Rhat Bulk_ESS Tail_ESS
sd(Intercept)   0.9964    0.0191   0.9584   1.0344 1.0003     1089     1988

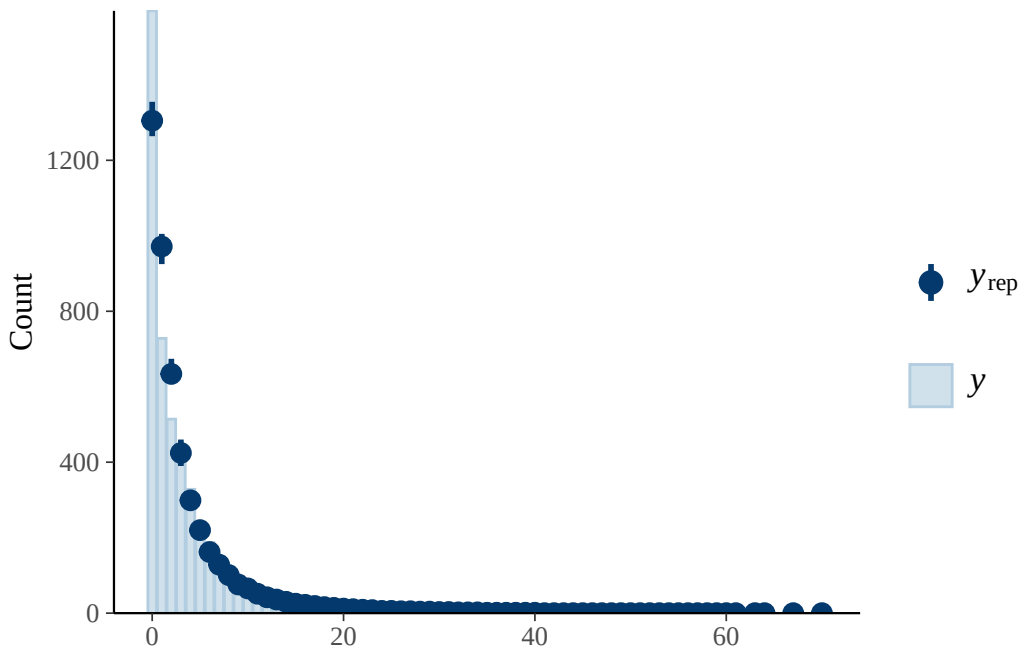
~site_id (Number of levels: 22)
              Estimate Est.Error l-95% CI u-95% CI   Rhat Bulk_ESS Tail_ESS
sd(Intercept)   0.0880    0.0376   0.0199   0.1681 1.0013      379      512

Regression Coefficients:
                                          Estimate Est.Error l-95% CI u-95% CI
Intercept                                   0.4521    0.0419   0.3668   0.5348
menarche_status_p_wave_2Y                   0.1591    0.0426   0.0759   0.2456
adhd_diagnosis1                             0.4666    0.0489   0.3720   0.5635
age_years_c_wave_2                         -0.0608    0.0279  -0.1163  -0.0072
ethnicity2                                 -0.1242    0.0663  -0.2574   0.0016
ethnicity3                                 -0.0821    0.0566  -0.1951   0.0273
ethnicity4                                 -0.4346    0.1408  -0.7104  -0.1630
ethnicity5                                  0.0379    0.0646  -0.0889   0.1634
inr_c_wave_0                               -0.0021    0.0066  -0.0148   0.0112
cbcl_scr_syn_external_r_c_wave_0            0.0962    0.0032   0.0898   0.1024
menarche_status_p_wave_2Y:adhd_diagnosis1   0.0292    0.0733  -0.1127   0.1748
                                            Rhat Bulk_ESS Tail_ESS
Intercept                                 1.0015     1274     2238
menarche_status_p_wave_2Y                 1.0008     1321     2358
adhd_diagnosis1                           1.0007     1173     2278
age_years_c_wave_2                        1.0007     1213     2032
ethnicity2                                1.0018      771     1229
ethnicity3                                1.0022      944     1380
ethnicity4                                1.0004     1636     2618
```

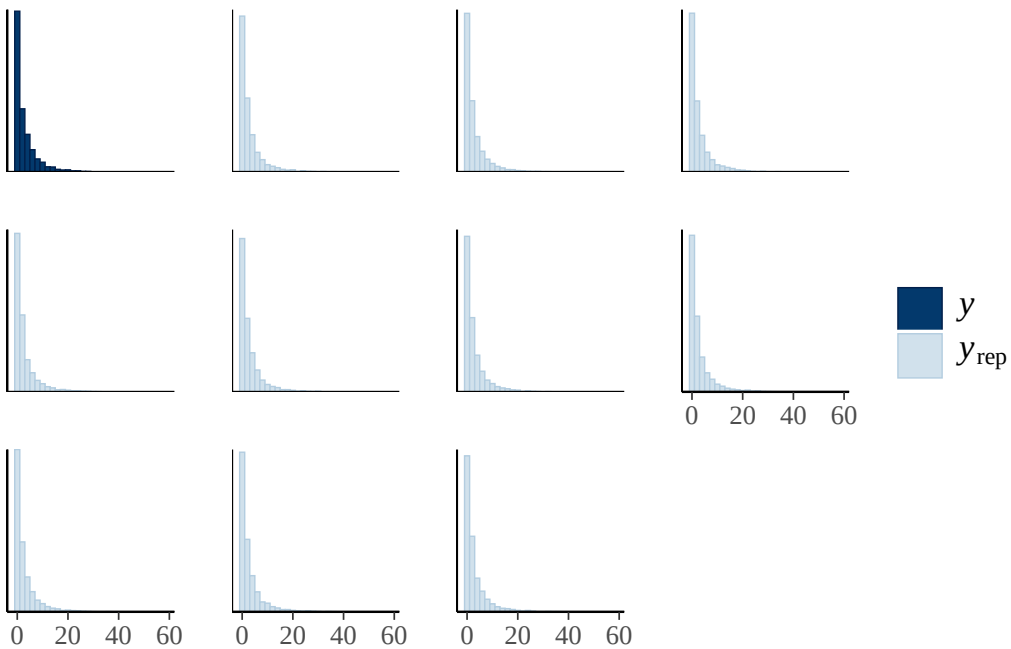
```
ethnicity5                                1.0015      891     1797
inr_c_wave_0                              1.0006      994     2063
cbcl_scr_syn_external_r_c_wave_0          1.0074      850     1770
menarche_status_p_wave_2Y:adhd_diagnosis1 1.0008      984     1506

Draws were sampled using sampling(NUTS). For each parameter, Bulk_ESS
and Tail_ESS are effective sample size measures, and Rhat is the potential
scale reduction factor on split chains (at convergence, Rhat = 1).
```

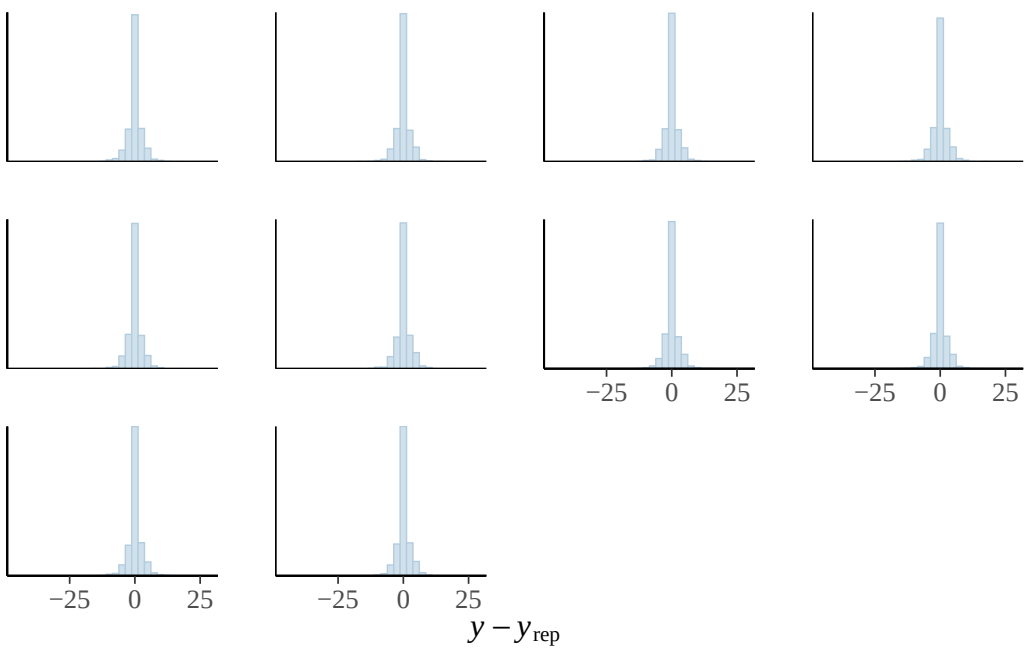
Model 3 long posterior predictive diagnostics



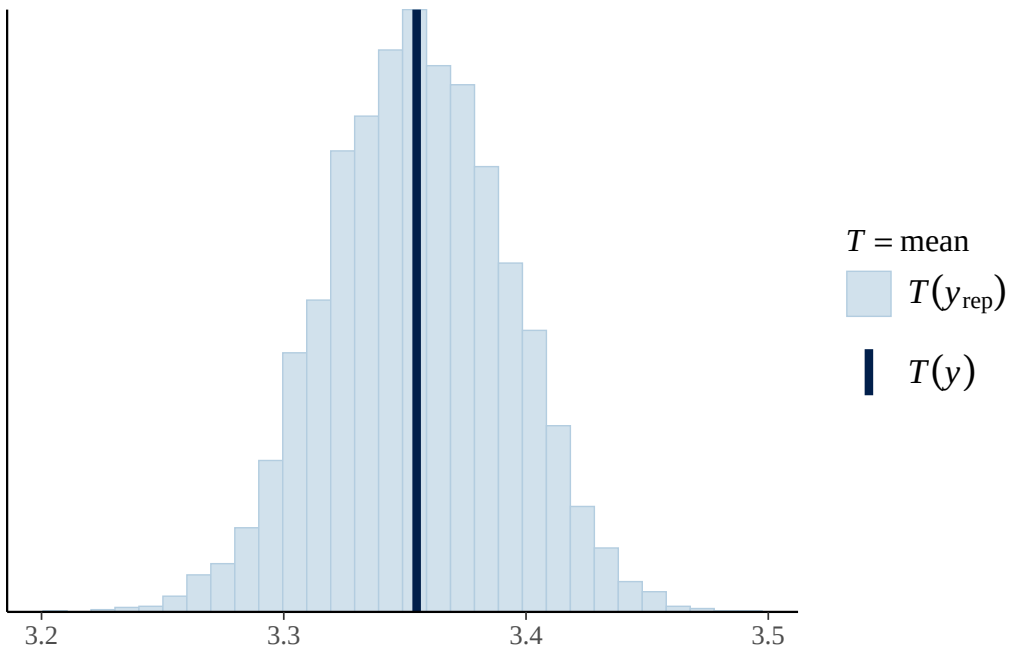

```
`stat_bin()` using `bins = 30`. Pick better value with `binwidth`.
```



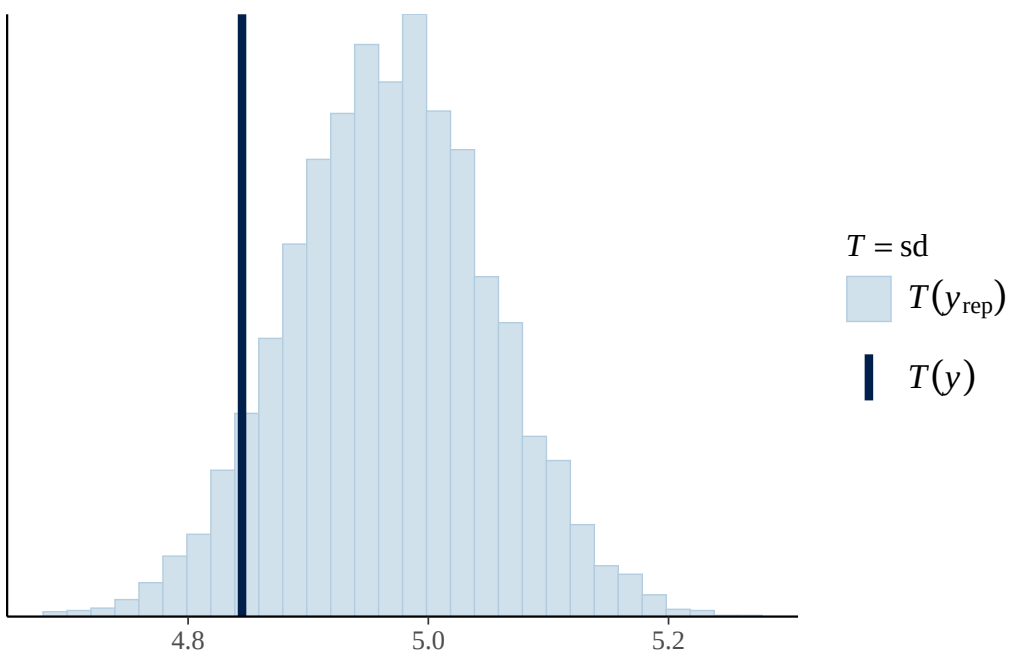

```
`stat_bin()` using `bins = 30`. Pick better value with `binwidth`.
```

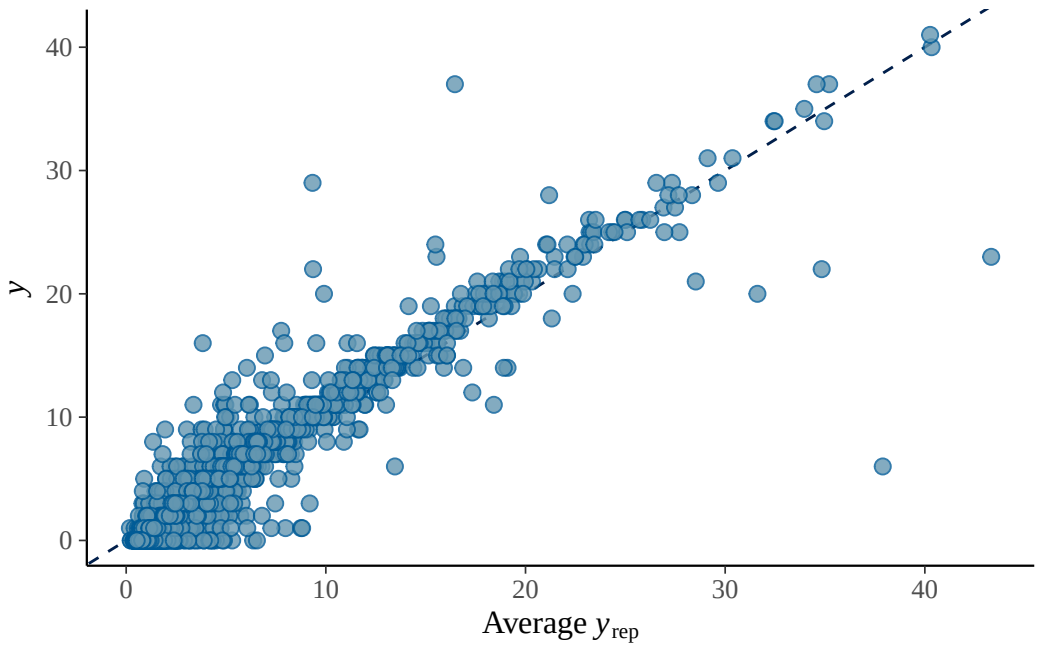



```
`stat_bin()` using `bins = 30`. Pick better value with `binwidth`.
```




```
`stat_bin()` using `bins = 30`. Pick better value with `binwidth`.
```





Model 3b longitudinal output

```
   Estimate   Est.Error      Q2.5     Q97.5
R2 0.827564 0.004608388 0.8183025 0.8362426

Warning: There were 10 divergent transitions after warmup. Increasing
adapt_delta above 0.8 may help. See
http://mc-stan.org/misc/warnings.html#divergent-transitions-after-warmup

 Family: poisson
  Links: mu = log
Formula: cbcl_scr_syn_external_r_wave_3 ~ pds_f4_p_c_wave_2 * adhd_diagnosis + age_years_c_wa
   Data: imp_df_wide (Number of observations: 4820)
  Draws: 2 chains, each with iter = 5000; warmup = 2500; thin = 1;
         total post-warmup draws = 5000

Multilevel Hyperparameters:
~family_id (Number of levels: 4177)
              Estimate Est.Error l-95% CI u-95% CI   Rhat Bulk_ESS Tail_ESS
sd(Intercept)   0.9981    0.0194   0.9614   1.0366 1.0007      973     1637

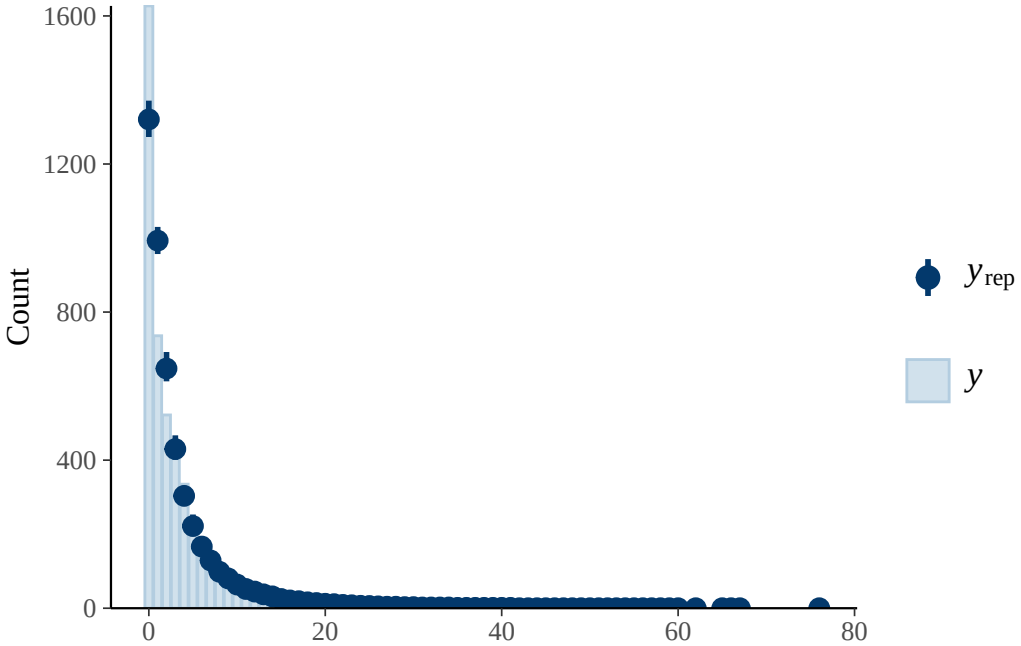
~site_id (Number of levels: 22)
```

```
              Estimate Est.Error l-95% CI u-95% CI   Rhat Bulk_ESS Tail_ESS
sd(Intercept)   0.0854    0.0394   0.0088   0.1631 1.0041      295      369

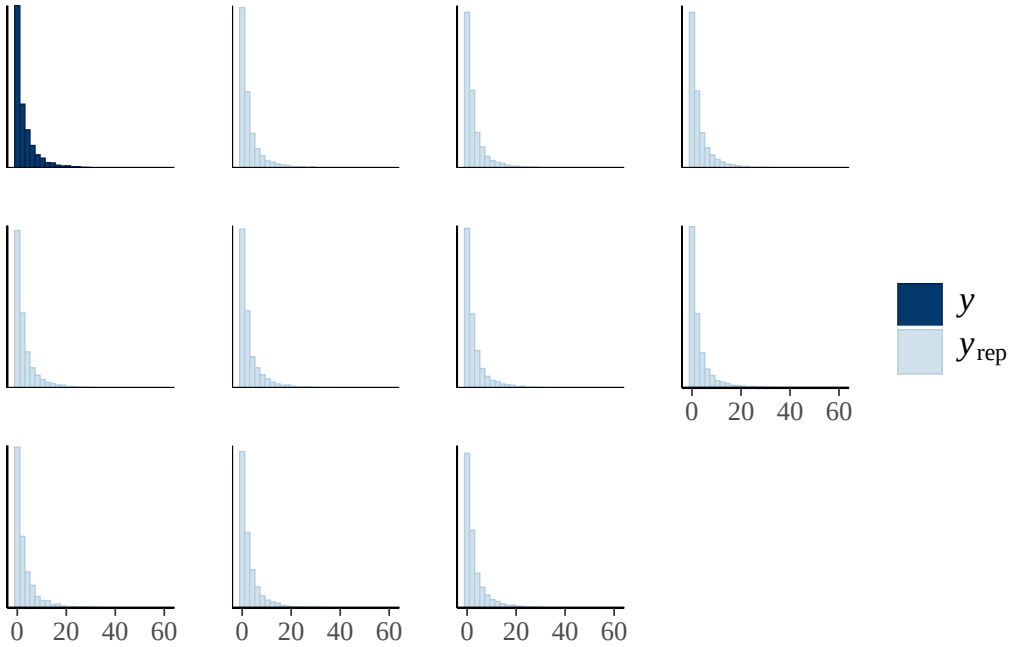
Regression Coefficients:
                                 Estimate Est.Error l-95% CI u-95% CI   Rhat
Intercept                          0.4786    0.0401   0.3984   0.5560 1.0002
pds_f4_p_c_wave_2                  0.1624    0.0318   0.0990   0.2253 1.0000
adhd_diagnosis1                    0.4717    0.0424   0.3881   0.5548 1.0012
age_years_c_wave_2                -0.0536    0.0272  -0.1063   0.0009 1.0003
ethnicity2                        -0.1220    0.0650  -0.2515   0.0037 0.9998
ethnicity3                        -0.0726    0.0568  -0.1818   0.0400 1.0012
ethnicity4                        -0.4506    0.1389  -0.7181  -0.1730 1.0004
ethnicity5                         0.0138    0.0623  -0.1101   0.1339 1.0034
inr_c_wave_0                      -0.0041    0.0063  -0.0167   0.0079 1.0011
cbcl_scr_syn_external_r_c_wave_0   0.0944    0.0031   0.0885   0.1004 1.0032
pds_f4_p_c_wave_2:adhd_diagnosis1 -0.0392    0.0577  -0.1522   0.0692 1.0004
                                 Bulk_ESS Tail_ESS
Intercept                            1587     2095
pds_f4_p_c_wave_2                    1495     2132
adhd_diagnosis1                      1315     1983
age_years_c_wave_2                   1399     2328
ethnicity2                            801     1940
ethnicity3                           1104     1825
ethnicity4                           1540     3026
ethnicity5                            965     1858
inr_c_wave_0                         1169     2228
cbcl_scr_syn_external_r_c_wave_0      898     1681
pds_f4_p_c_wave_2:adhd_diagnosis1    1262     2375

Draws were sampled using sampling(NUTS). For each parameter, Bulk_ESS
and Tail_ESS are effective sample size measures, and Rhat is the potential
scale reduction factor on split chains (at convergence, Rhat = 1).
```

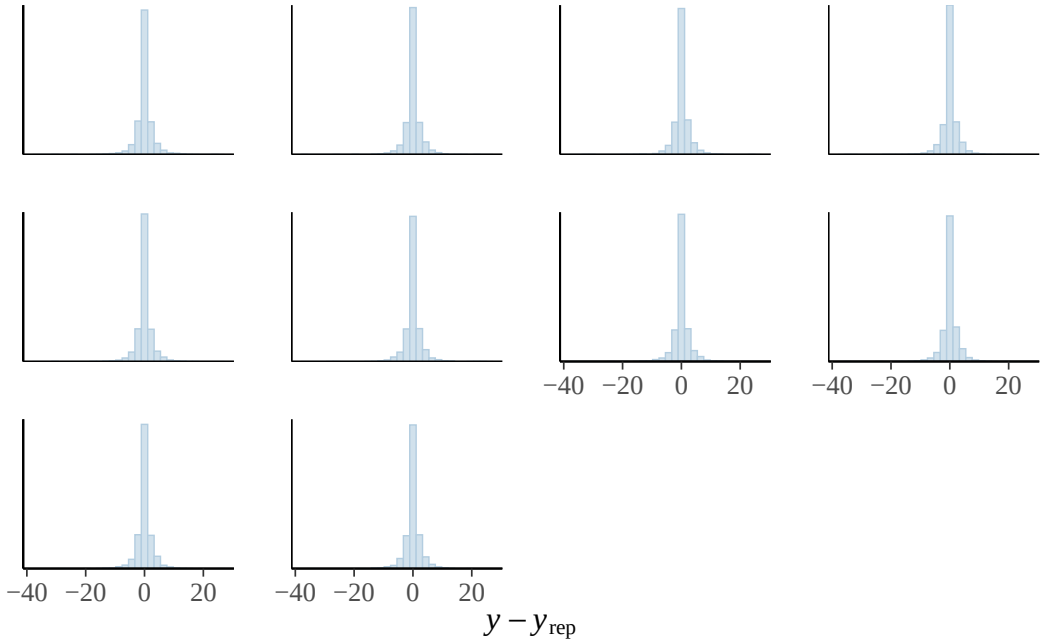
Model 3b long posterior predictive diagnostics



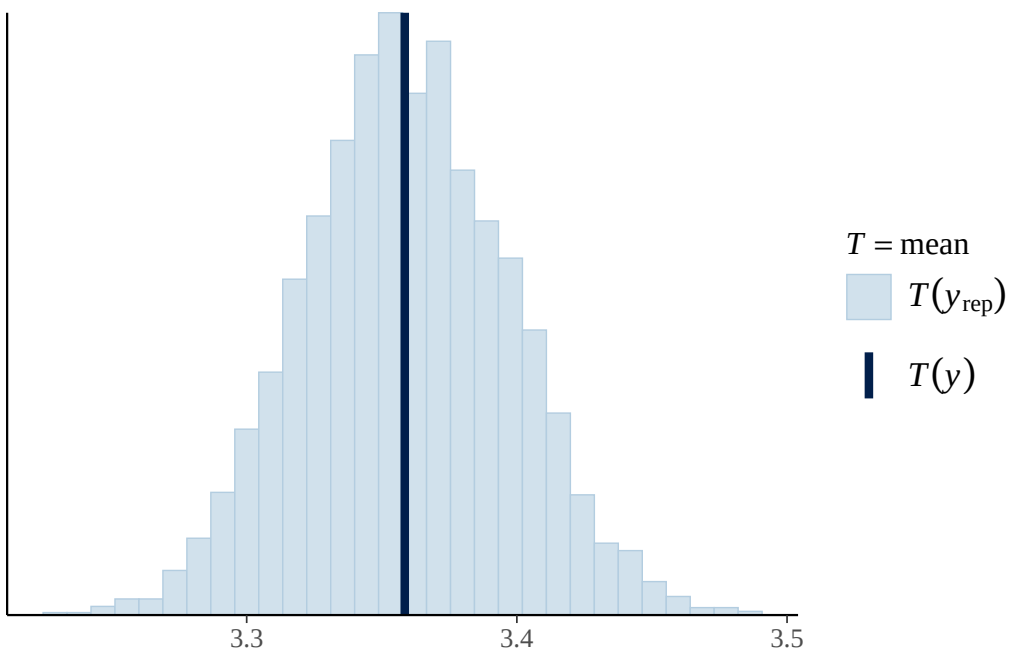

```
`stat_bin()` using `bins = 30`. Pick better value with `binwidth`.
```



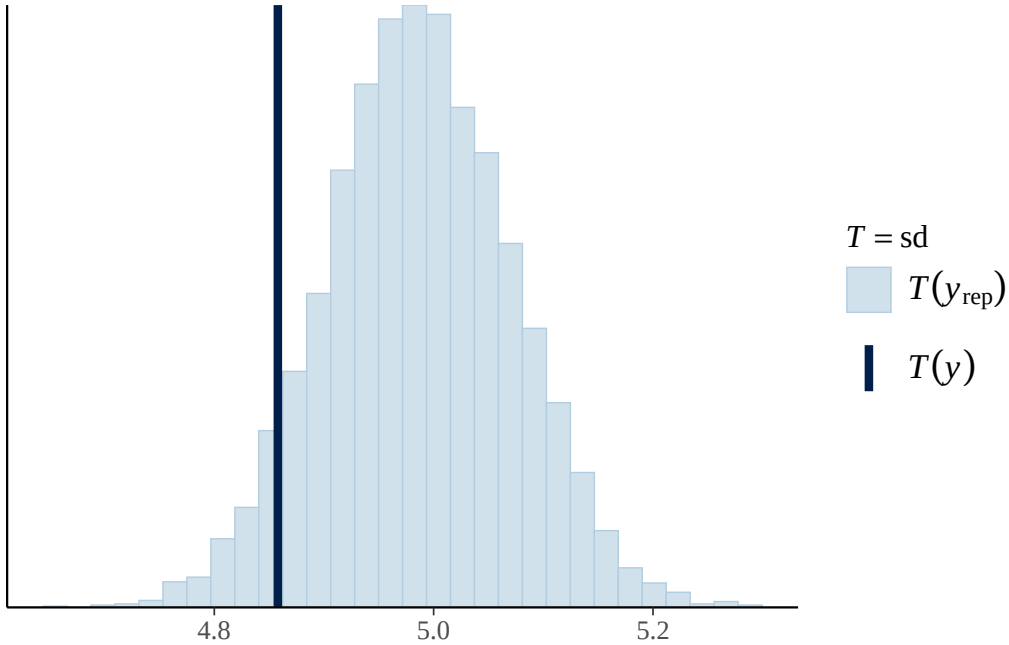
`stat_bin()` using `bins = 30`. Pick better value with `binwidth`.

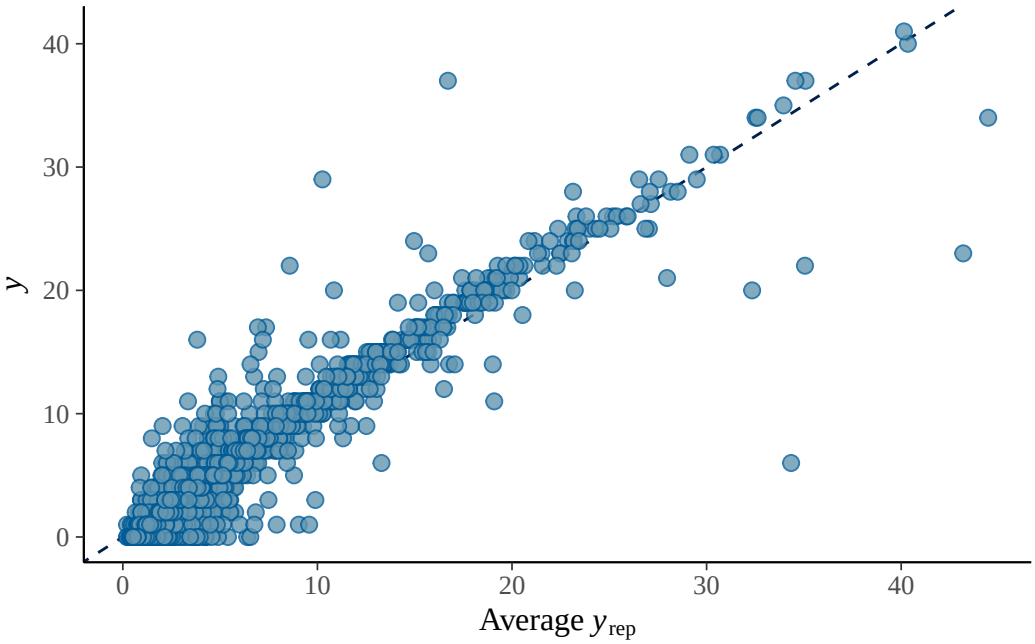


`stat_bin()` using `bins = 30`. Pick better value with `binwidth`.



```
`stat_bin()` using `bins = 30`. Pick better value with `binwidth`.
```





Model 4 longitudinal output

```
    Estimate   Est.Error      Q2.5     Q97.5
R2 0.8368844 0.004163148 0.8285755 0.8448409


 Family: poisson
  Links: mu = log
Formula: cbcl_scr_syn_internal_r_wave_3 ~ menarche_status_p_wave_2 * cbcl_scr_syn_attention_r
   Data: imp_df_wide (Number of observations: 4740)
  Draws: 2 chains, each with iter = 5000; warmup = 2500; thin = 1;
         total post-warmup draws = 5000

Multilevel Hyperparameters:
~family_id (Number of levels: 4111)
              Estimate Est.Error l-95% CI u-95% CI   Rhat Bulk_ESS Tail_ESS
sd(Intercept)   0.8046    0.0140   0.7772   0.8313 1.0009     1305     2313

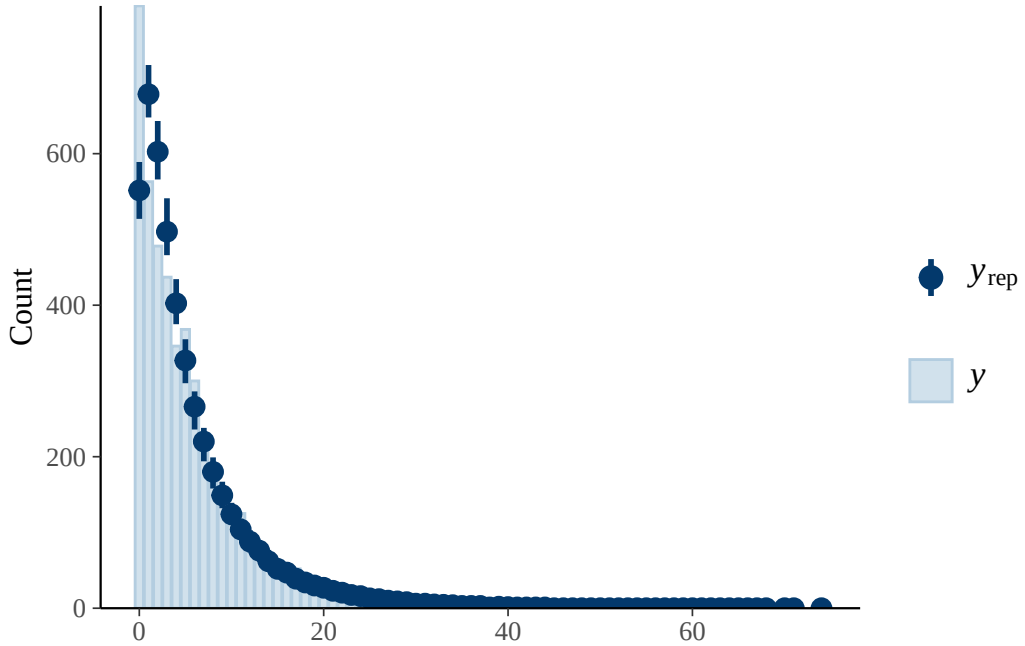
~site_id (Number of levels: 22)
              Estimate Est.Error l-95% CI u-95% CI   Rhat Bulk_ESS Tail_ESS
sd(Intercept)   0.1088    0.0294   0.0580   0.1720 1.0010      735     1047

Regression Coefficients:
                                                          Estimate Est.Error
Intercept                                                   1.3104    0.0357
menarche_status_p_wave_2Y                                   0.1816    0.0301
cbcl_scr_syn_attention_r_c_wave_2                           0.0720    0.0056
age_years_c_wave_2                                          0.0003    0.0217
ethnicity2                                                 -0.4161    0.0528
ethnicity3                                                 -0.0700    0.0464
ethnicity4                                                 -0.3468    0.1052
ethnicity5                                                  0.0738    0.0487
inr_c_wave_0                                                0.0168    0.0049
cbcl_scr_syn_internal_r_c_wave_0                            0.0705    0.0024
menarche_status_p_wave_2Y:cbcl_scr_syn_attention_r_c_wave_2  0.0003    0.0081
                                                          l-95% CI u-95% CI
Intercept                                                   1.2398   1.3791
menarche_status_p_wave_2Y                                   0.1219   0.2402
cbcl_scr_syn_attention_r_c_wave_2                           0.0609   0.0832
age_years_c_wave_2                                         -0.0413   0.0435
ethnicity2                                                 -0.5169  -0.3140
ethnicity3                                                 -0.1583   0.0241
ethnicity4                                                 -0.5575  -0.1476
```

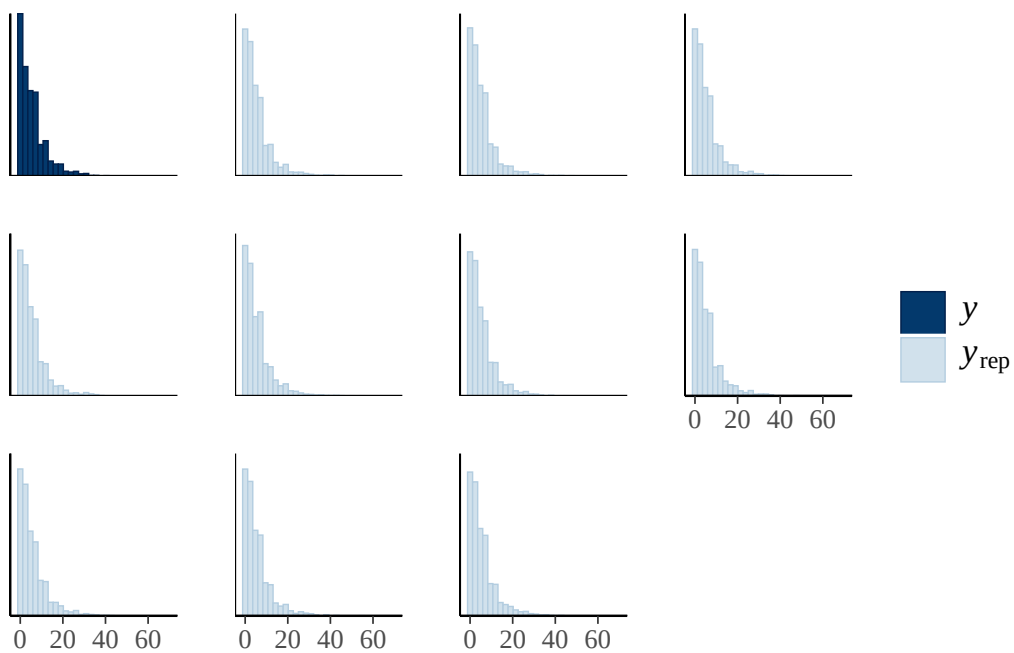
```
ethnicity5                                                  -0.0220   0.1665
inr_c_wave_0                                                 0.0067   0.0261
cbcl_scr_syn_internal_r_c_wave_0                             0.0658   0.0752
menarche_status_p_wave_2Y:cbcl_scr_syn_attention_r_c_wave_2 -0.0155   0.0160
                                                             Rhat Bulk_ESS
Intercept                                                  1.0000     1546
menarche_status_p_wave_2Y                                  1.0026     1258
cbcl_scr_syn_attention_r_c_wave_2                          1.0049     1221
age_years_c_wave_2                                         1.0036     1434
ethnicity2                                                 1.0004     1468
ethnicity3                                                 1.0011     1174
ethnicity4                                                 1.0002     2172
ethnicity5                                                 0.9999     1221
inr_c_wave_0                                               1.0001     1481
cbcl_scr_syn_internal_r_c_wave_0                           1.0000     1120
menarche_status_p_wave_2Y:cbcl_scr_syn_attention_r_c_wave_2 1.0068     1292
                                                           Tail_ESS
Intercept                                                      2418
menarche_status_p_wave_2Y                                      2436
cbcl_scr_syn_attention_r_c_wave_2                              1819
age_years_c_wave_2                                             2554
ethnicity2                                                     2552
ethnicity3                                                     2070
ethnicity4                                                     3158
ethnicity5                                                     2078
inr_c_wave_0                                                   1873
cbcl_scr_syn_internal_r_c_wave_0                               2205
menarche_status_p_wave_2Y:cbcl_scr_syn_attention_r_c_wave_2     1981

Draws were sampled using sampling(NUTS). For each parameter, Bulk_ESS
and Tail_ESS are effective sample size measures, and Rhat is the potential
scale reduction factor on split chains (at convergence, Rhat = 1).
```

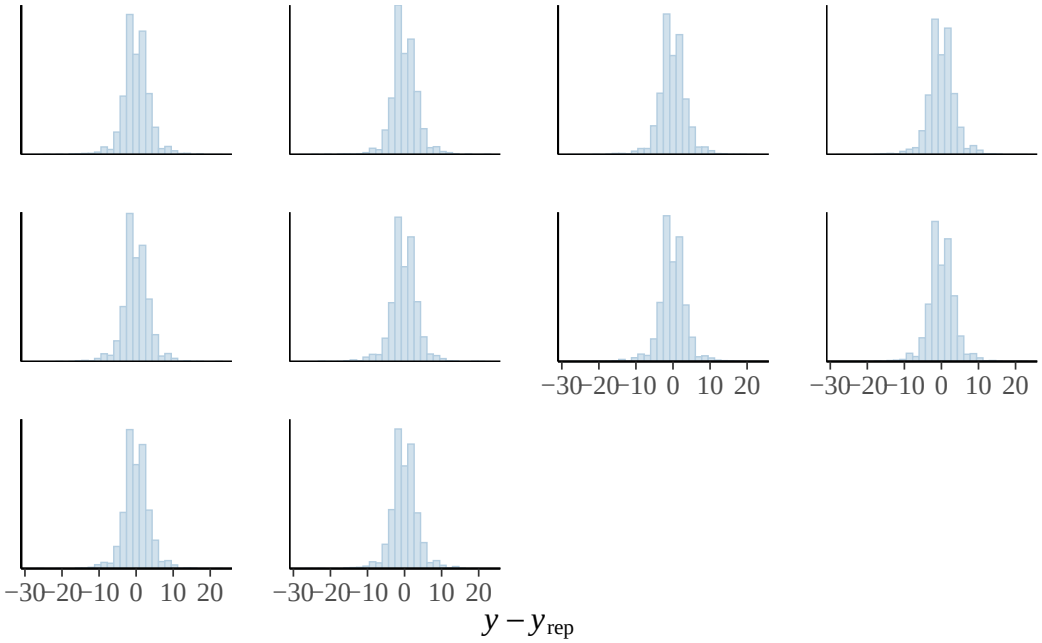
Model 4 long posterior predictive diagnostics



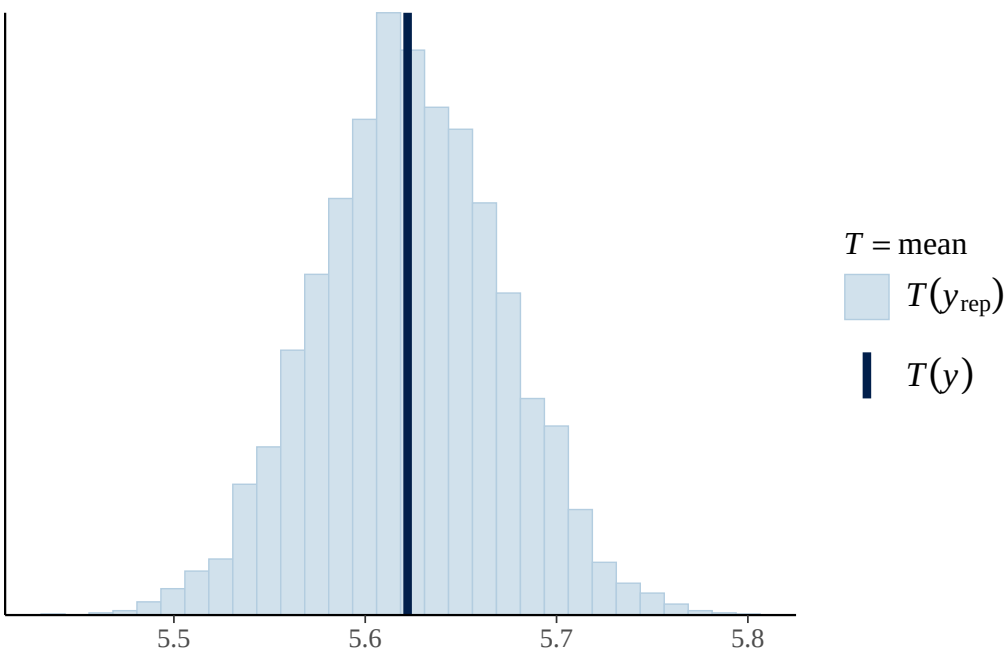

```
`stat_bin()` using `bins = 30`. Pick better value with `binwidth`.
```



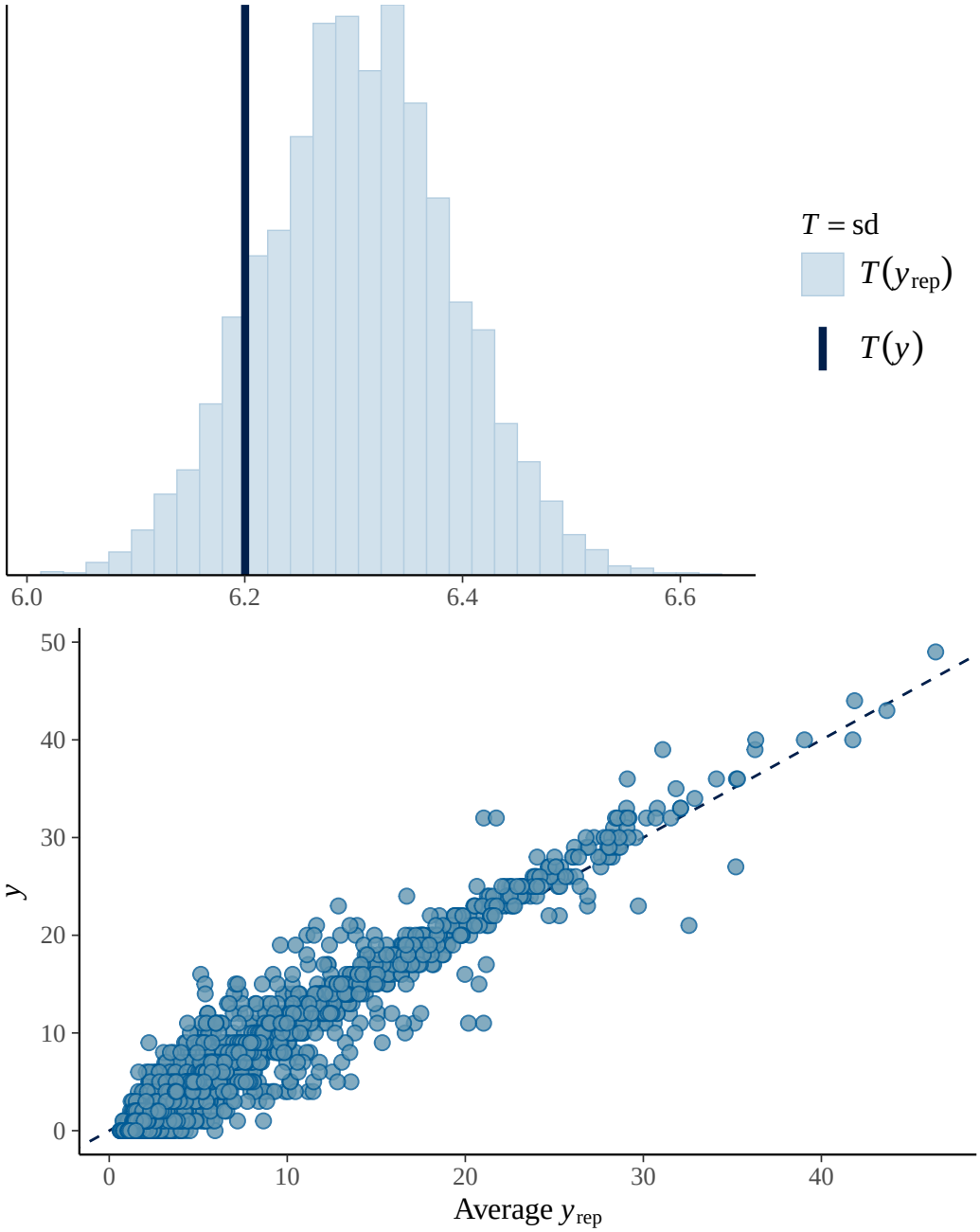
```
`stat_bin()` using `bins = 30`. Pick better value with `binwidth`.
```




```
`stat_bin()` using `bins = 30`. Pick better value with `binwidth`.
```



```
`stat_bin()` using `bins = 30`. Pick better value with `binwidth`.
```



Model 4b longitudinal output

```
    Estimate   Est.Error     Q2.5     Q97.5
R2 0.8369294 0.004185831 0.828679 0.8449466


 Family: poisson
  Links: mu = log
Formula: cbcl_scr_syn_internal_r_wave_3 ~ pds_f4_p_c_wave_2 * cbcl_scr_syn_attention_r_c_wave
   Data: imp_df_wide (Number of observations: 4820)
  Draws: 2 chains, each with iter = 5000; warmup = 2500; thin = 1;
         total post-warmup draws = 5000

Multilevel Hyperparameters:
~family_id (Number of levels: 4177)
              Estimate Est.Error l-95% CI u-95% CI   Rhat Bulk_ESS Tail_ESS
sd(Intercept)   0.8121    0.0141   0.7852   0.8409 1.0021     1176     2183

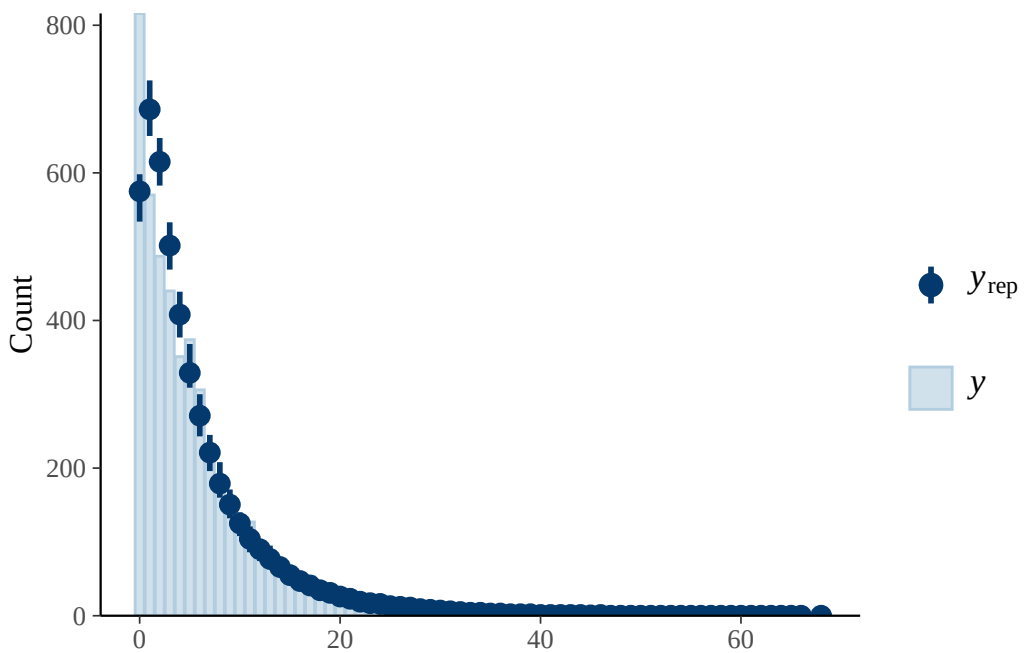
~site_id (Number of levels: 22)
              Estimate Est.Error l-95% CI u-95% CI   Rhat Bulk_ESS Tail_ESS
sd(Intercept)   0.1105    0.0290   0.0621   0.1748 1.0019      877     1582

Regression Coefficients:
                                                   Estimate Est.Error l-95% CI
Intercept                                            1.3341    0.0358   1.2613
pds_f4_p_c_wave_2                                    0.1229    0.0215   0.0813
cbcl_scr_syn_attention_r_c_wave_2                    0.0658    0.0045   0.0569
age_years_c_wave_2                                   0.0230    0.0206  -0.0176
ethnicity2                                          -0.3986    0.0516  -0.4974
ethnicity3                                          -0.0538    0.0453  -0.1409
ethnicity4                                          -0.3457    0.1082  -0.5587
ethnicity5                                           0.0563    0.0488  -0.0390
inr_c_wave_0                                         0.0154    0.0050   0.0057
cbcl_scr_syn_internal_r_c_wave_0                     0.0709    0.0024   0.0663
pds_f4_p_c_wave_2:cbcl_scr_syn_attention_r_c_wave_2  0.0104    0.0062  -0.0018
                                                   u-95% CI   Rhat Bulk_ESS
Intercept                                            1.4032 1.0042     1512
pds_f4_p_c_wave_2                                    0.1666 1.0003     1574
cbcl_scr_syn_attention_r_c_wave_2                    0.0746 1.0022     1707
age_years_c_wave_2                                   0.0625 0.9998     1728
ethnicity2                                          -0.2971 1.0051     1331
ethnicity3                                           0.0355 1.0017     1274
ethnicity4                                          -0.1326 1.0005     1846
```

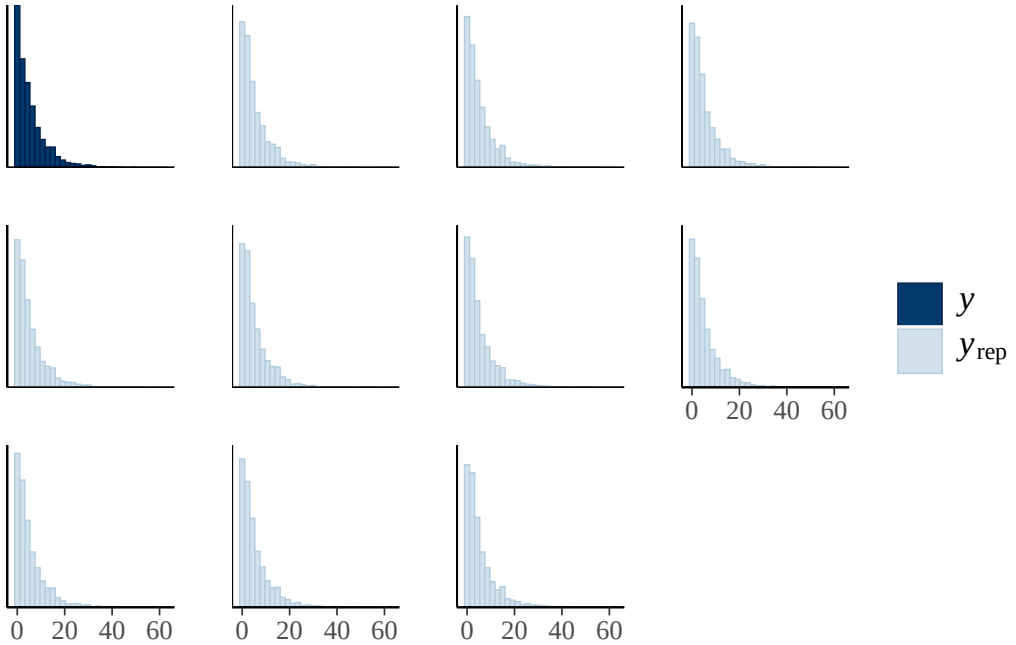
```
ethnicity5                                            0.1501 1.0027     1215
inr_c_wave_0                                          0.0255 1.0007     1279
cbcl_scr_syn_internal_r_c_wave_0                      0.0757 1.0019     1117
pds_f4_p_c_wave_2:cbcl_scr_syn_attention_r_c_wave_2   0.0225 1.0010     1687
                                                     Tail_ESS
Intercept                                               2319
pds_f4_p_c_wave_2                                       2834
cbcl_scr_syn_attention_r_c_wave_2                       2540
age_years_c_wave_2                                      2836
ethnicity2                                              2363
ethnicity3                                              2091
ethnicity4                                              3064
ethnicity5                                              2471
inr_c_wave_0                                            2007
cbcl_scr_syn_internal_r_c_wave_0                        2450
pds_f4_p_c_wave_2:cbcl_scr_syn_attention_r_c_wave_2     3079

Draws were sampled using sampling(NUTS). For each parameter, Bulk_ESS
and Tail_ESS are effective sample size measures, and Rhat is the potential
scale reduction factor on split chains (at convergence, Rhat = 1).
```

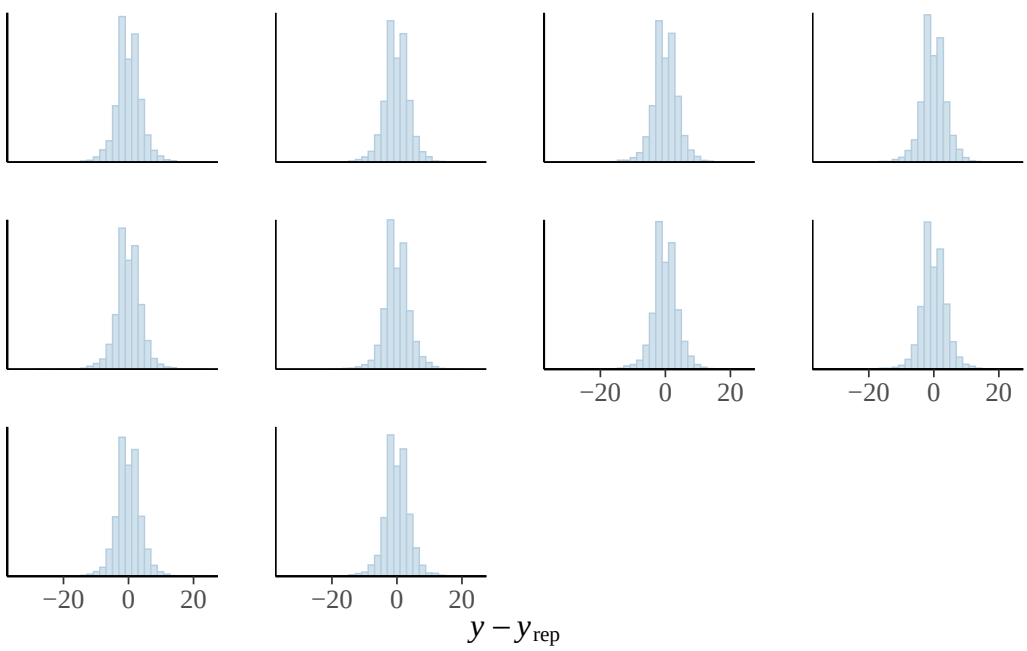
Model 4b long posterior predictive diagnostics



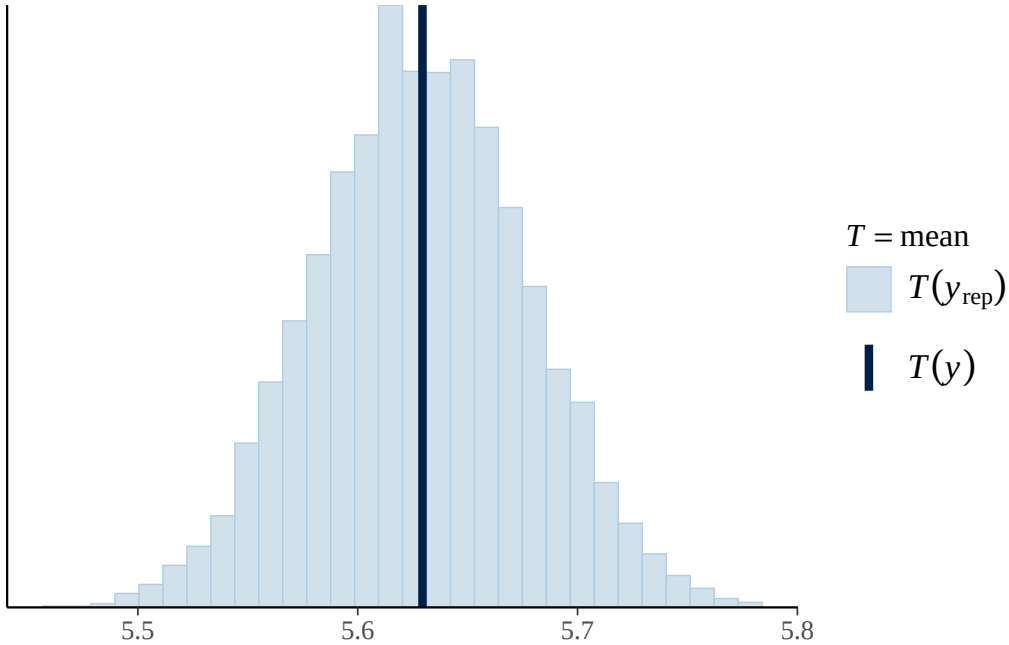
```
`stat_bin()` using `bins = 30`. Pick better value with `binwidth`.
```



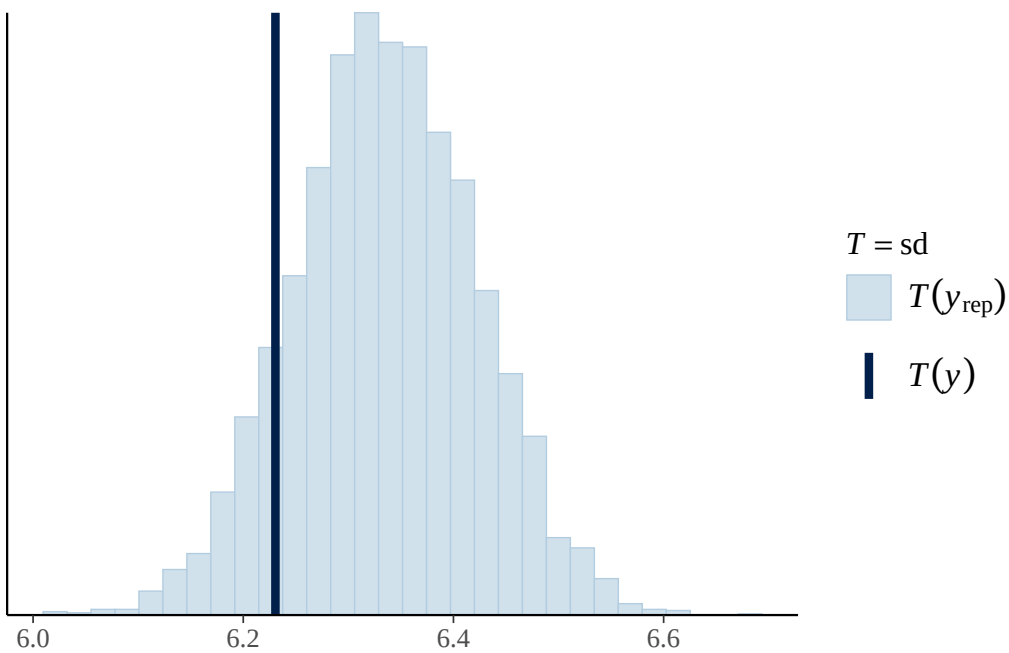

```
`stat_bin()` using `bins = 30`. Pick better value with `binwidth`.
```

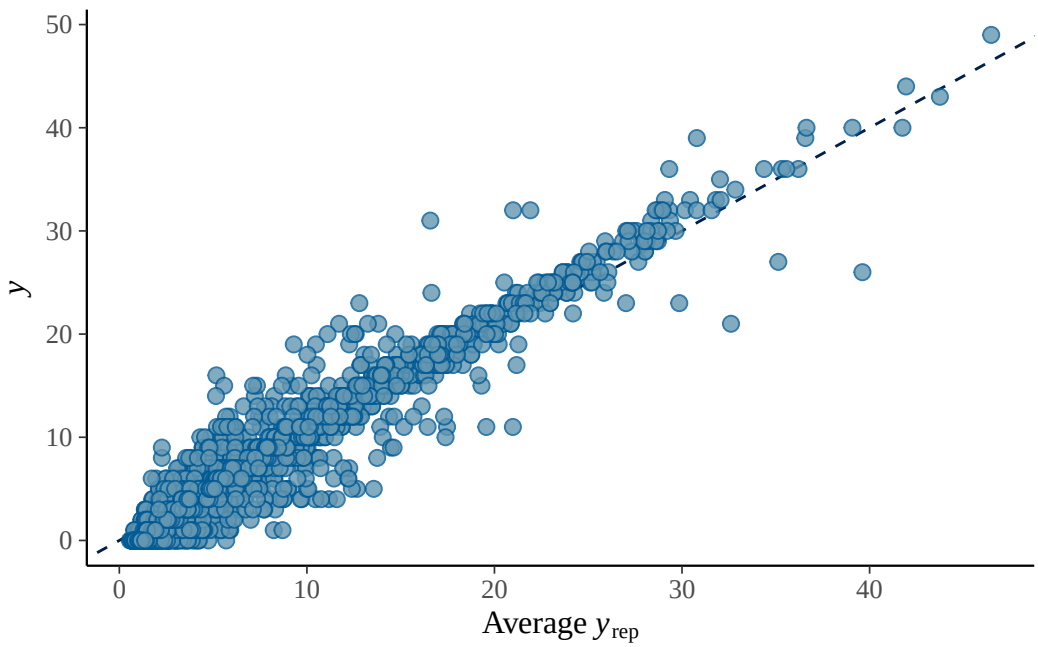


```
`stat_bin()` using `bins = 30`. Pick better value with `binwidth`.
```




```
`stat_bin()` using `bins = 30`. Pick better value with `binwidth`.
```





Model 5 longitudinal output

```
    Estimate Est.Error      Q2.5     Q97.5
R2 0.8275591 0.0047251 0.8181307 0.8366149


 Family: poisson
  Links: mu = log
Formula: cbcl_scr_syn_external_r_wave_3 ~ menarche_status_p_wave_2 * cbcl_scr_syn_attention_
   Data: imp_df_wide (Number of observations: 4740)
  Draws: 2 chains, each with iter = 5000; warmup = 2500; thin = 1;
         total post-warmup draws = 5000

Multilevel Hyperparameters:
~family_id (Number of levels: 4111)
              Estimate Est.Error l-95% CI u-95% CI   Rhat Bulk_ESS Tail_ESS
sd(Intercept)   0.9510    0.0191   0.9133   0.9885 1.0013     1025     1868

~site_id (Number of levels: 22)
              Estimate Est.Error l-95% CI u-95% CI   Rhat Bulk_ESS Tail_ESS
sd(Intercept)   0.0927    0.0367   0.0212   0.1691 1.0030      373      399

Regression Coefficients:
```

	Estimate	Est.Error
Intercept	0.5740	0.0401
menarche_status_p_wave_2Y	0.1278	0.0377
cbcl_scr_syn_attention_r_c_wave_2	0.0974	0.0068
age_years_c_wave_2	-0.0634	0.0281
ethnicity2	-0.1236	0.0626
ethnicity3	-0.0731	0.0556
ethnicity4	-0.3877	0.1367
ethnicity5	0.0057	0.0604
inr_c_wave_0	0.0005	0.0064
cbcl_scr_syn_external_r_c_wave_0	0.0880	0.0031
menarche_status_p_wave_2Y:cbcl_scr_syn_attention_r_c_wave_2	0.0126	0.0098

	l-95% CI	u-95% CI
Intercept	0.4959	0.6505
menarche_status_p_wave_2Y	0.0553	0.2028
cbcl_scr_syn_attention_r_c_wave_2	0.0844	0.1109
age_years_c_wave_2	-0.1202	-0.0078
ethnicity2	-0.2474	-0.0035
ethnicity3	-0.1831	0.0334
ethnicity4	-0.6585	-0.1267
ethnicity5	-0.1124	0.1273
inr_c_wave_0	-0.0122	0.0128
cbcl_scr_syn_external_r_c_wave_0	0.0819	0.0940
menarche_status_p_wave_2Y:cbcl_scr_syn_attention_r_c_wave_2	-0.0067	0.0318

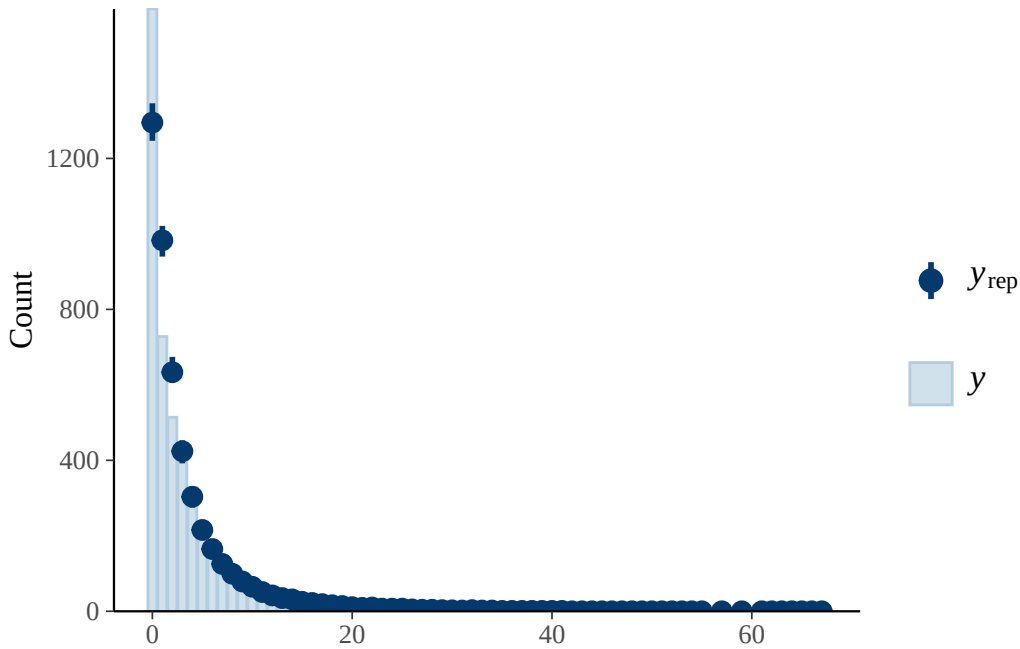
	Rhat	Bulk_ESS
Intercept	1.0032	1401
menarche_status_p_wave_2Y	1.0010	1560
cbcl_scr_syn_attention_r_c_wave_2	1.0005	1205
age_years_c_wave_2	1.0037	1240
ethnicity2	1.0067	851
ethnicity3	1.0031	1028
ethnicity4	1.0006	1812
ethnicity5	1.0033	1097
inr_c_wave_0	1.0019	1100
cbcl_scr_syn_external_r_c_wave_0	1.0003	891
menarche_status_p_wave_2Y:cbcl_scr_syn_attention_r_c_wave_2	1.0000	1104

	Tail_ESS
Intercept	2331
menarche_status_p_wave_2Y	2807
cbcl_scr_syn_attention_r_c_wave_2	2121
age_years_c_wave_2	2024
ethnicity2	1647
ethnicity3	2238

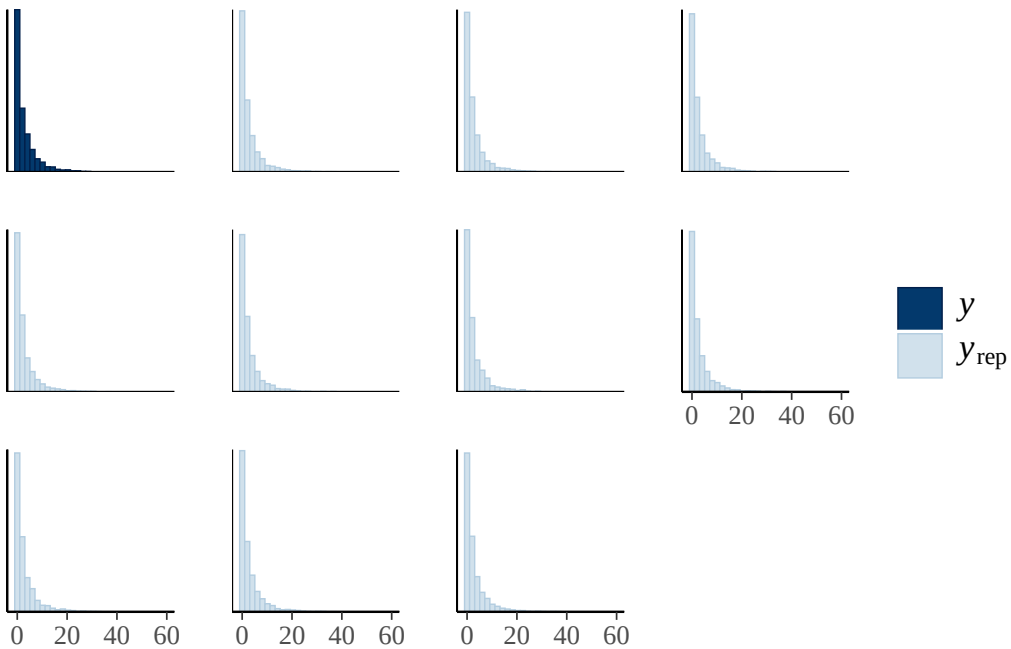
```
ethnicity4                                                        2732
ethnicity5                                                        1751
inr_c_wave_0                                                      1949
cbcl_scr_syn_external_r_c_wave_0                                  1862
menarche_status_p_wave_2Y:cbcl_scr_syn_attention_r_c_wave_2       1662

Draws were sampled using sampling(NUTS). For each parameter, Bulk_ESS
and Tail_ESS are effective sample size measures, and Rhat is the potential
scale reduction factor on split chains (at convergence, Rhat = 1).
```

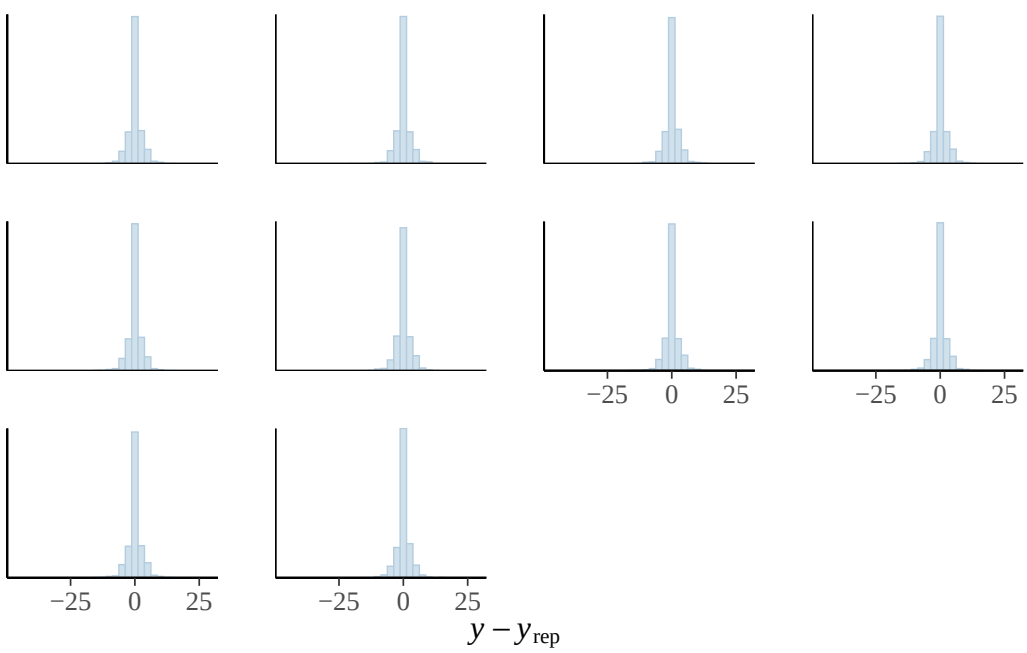
Model 5 long posterior predictive diagnostics



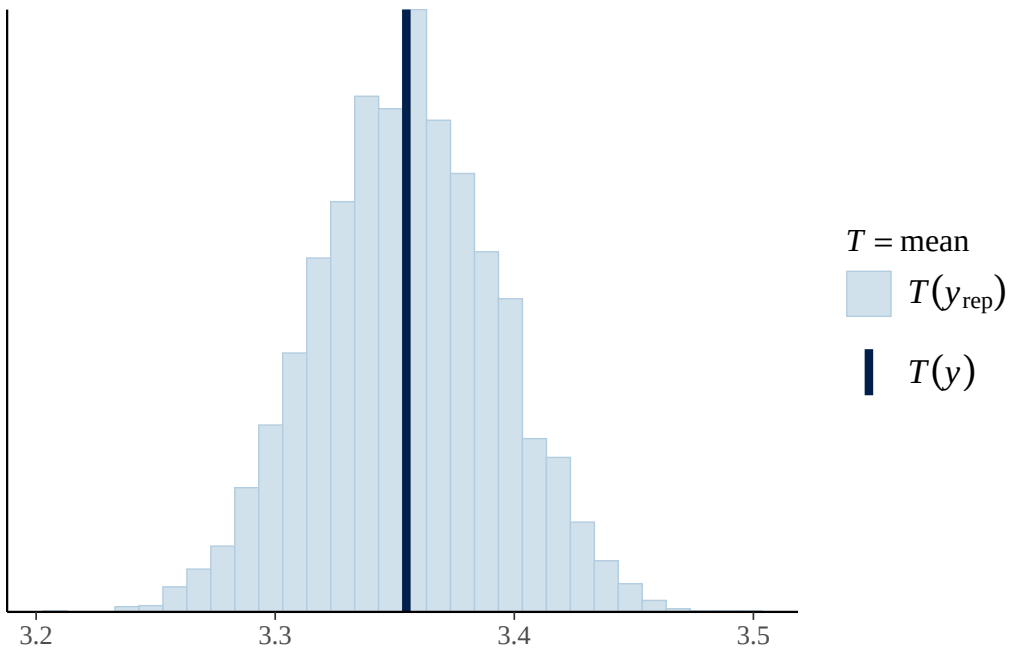

```
`stat_bin()` using `bins = 30`. Pick better value with `binwidth`.
```



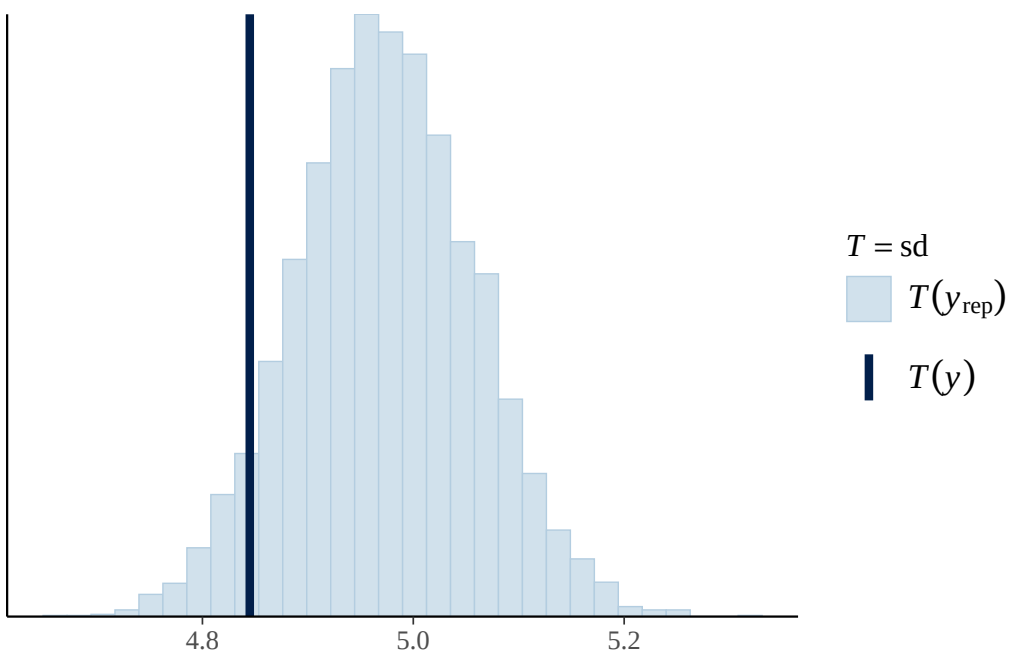

```
`stat_bin()` using `bins = 30`. Pick better value with `binwidth`.
```

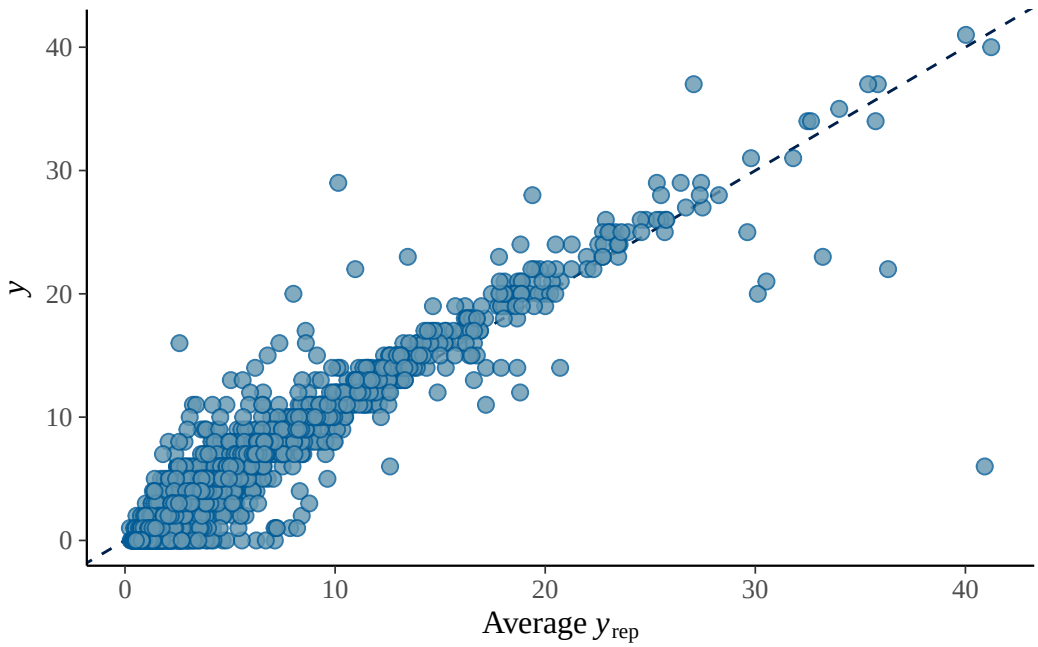



```
`stat_bin()` using `bins = 30`. Pick better value with `binwidth`.
```




```
`stat_bin()` using `bins = 30`. Pick better value with `binwidth`.
```





Model 5b longitudinal output

```
    Estimate   Est.Error      Q2.5     Q97.5
R2 0.8118904 0.005174597 0.8013317 0.8215892


 Family: poisson
  Links: mu = log
Formula: cbcl_scr_syn_external_r_wave_3 ~ pds_f4_p_c_wave_2 * cbcl_scr_syn_external_r_c_wave_
   Data: imp_df_wide (Number of observations: 4820)
  Draws: 2 chains, each with iter = 5000; warmup = 2500; thin = 1;
         total post-warmup draws = 5000

Multilevel Hyperparameters:
~family_id (Number of levels: 4177)
              Estimate Est.Error l-95% CI u-95% CI   Rhat Bulk_ESS Tail_ESS
sd(Intercept)   0.8906    0.0179   0.8566   0.9262 1.0012     1173     2466

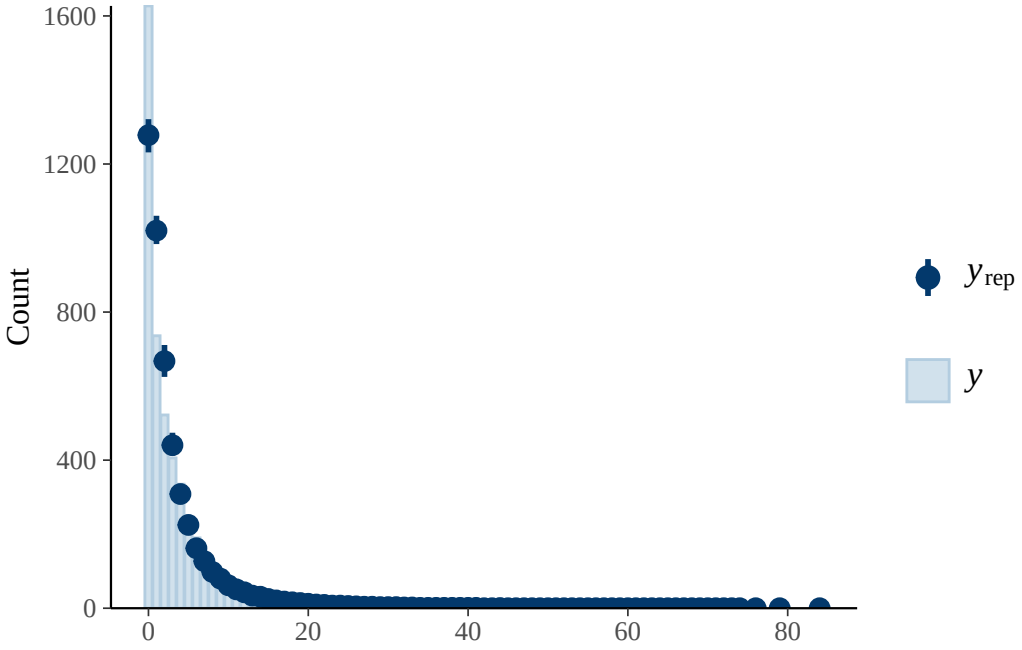
~site_id (Number of levels: 22)
              Estimate Est.Error l-95% CI u-95% CI   Rhat Bulk_ESS Tail_ESS
sd(Intercept)   0.0900    0.0353   0.0180   0.1639 1.0062      387      369

Regression Coefficients:
```

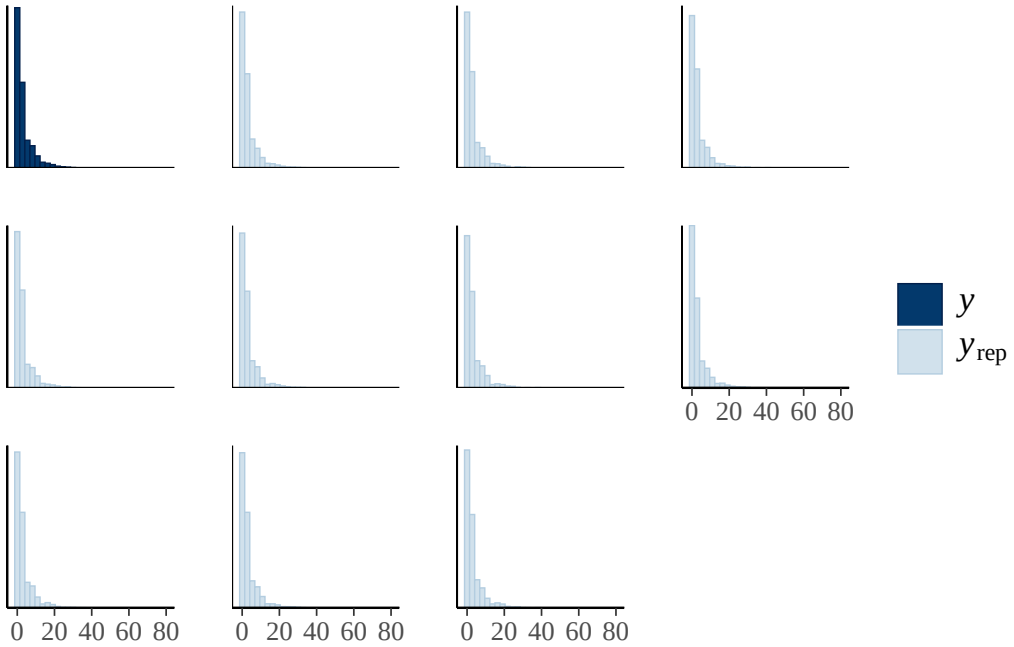
```
                                                Estimate Est.Error l-95% CI
Intercept                                         0.6313    0.0376   0.5534
pds_f4_p_c_wave_2                                 0.0979    0.0273   0.0449
cbcl_scr_syn_external_r_c_wave_2                  0.0736    0.0036   0.0663
age_years_c_wave_2                               -0.0580    0.0257  -0.1074
ethnicity2                                       -0.1135    0.0596  -0.2302
ethnicity3                                       -0.0908    0.0537  -0.1951
ethnicity4                                       -0.4165    0.1297  -0.6727
ethnicity5                                       -0.0132    0.0564  -0.1250
inr_c_wave_0                                      0.0012    0.0060  -0.0107
cbcl_scr_syn_external_r_c_wave_0                  0.0563    0.0036   0.0494
pds_f4_p_c_wave_2:cbcl_scr_syn_external_r_c_wave_2  0.0210    0.0039   0.0136
                                                u-95% CI   Rhat Bulk_ESS
Intercept                                         0.7033 1.0007     1063
pds_f4_p_c_wave_2                                 0.1525 1.0004     1531
cbcl_scr_syn_external_r_c_wave_2                  0.0806 1.0013     1044
age_years_c_wave_2                               -0.0069 1.0025     1225
ethnicity2                                        0.0055 1.0039      943
ethnicity3                                        0.0143 1.0045      868
ethnicity4                                       -0.1640 0.9998     1680
ethnicity5                                        0.0961 1.0008     1036
inr_c_wave_0                                      0.0129 1.0012     1026
cbcl_scr_syn_external_r_c_wave_0                  0.0634 1.0014      991
pds_f4_p_c_wave_2:cbcl_scr_syn_external_r_c_wave_2  0.0285 1.0019     1036
                                                Tail_ESS
Intercept                                           1868
pds_f4_p_c_wave_2                                   2702
cbcl_scr_syn_external_r_c_wave_2                    1823
age_years_c_wave_2                                  2133
ethnicity2                                          1610
ethnicity3                                          1888
ethnicity4                                          2565
ethnicity5                                          2363
inr_c_wave_0                                        1979
cbcl_scr_syn_external_r_c_wave_0                    1594
pds_f4_p_c_wave_2:cbcl_scr_syn_external_r_c_wave_2     1603

Draws were sampled using sampling(NUTS). For each parameter, Bulk_ESS
and Tail_ESS are effective sample size measures, and Rhat is the potential
scale reduction factor on split chains (at convergence, Rhat = 1).
```

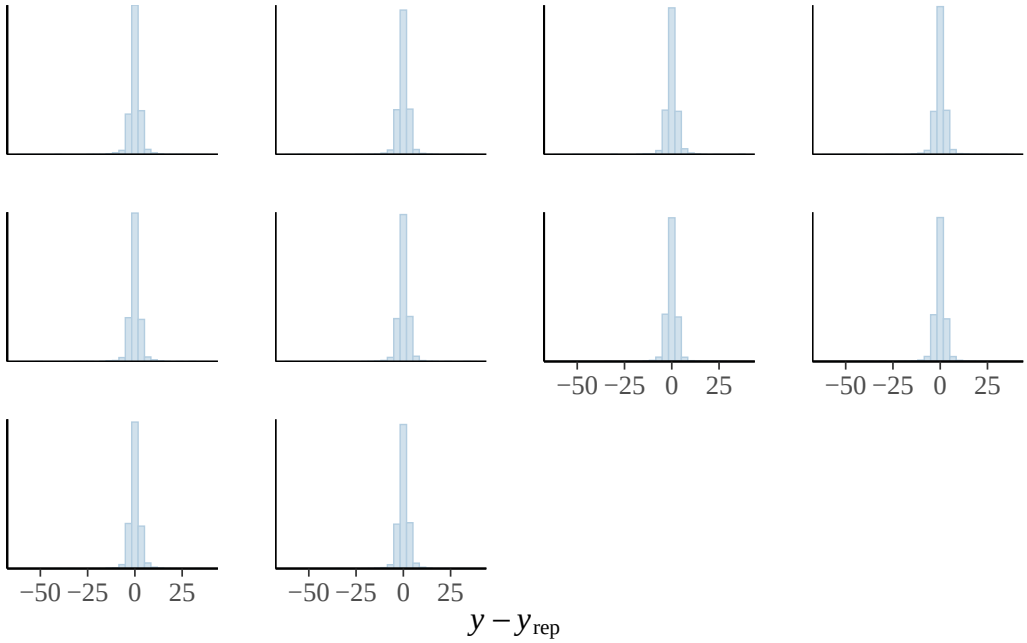
Model 5b long posterior predictive diagnostics



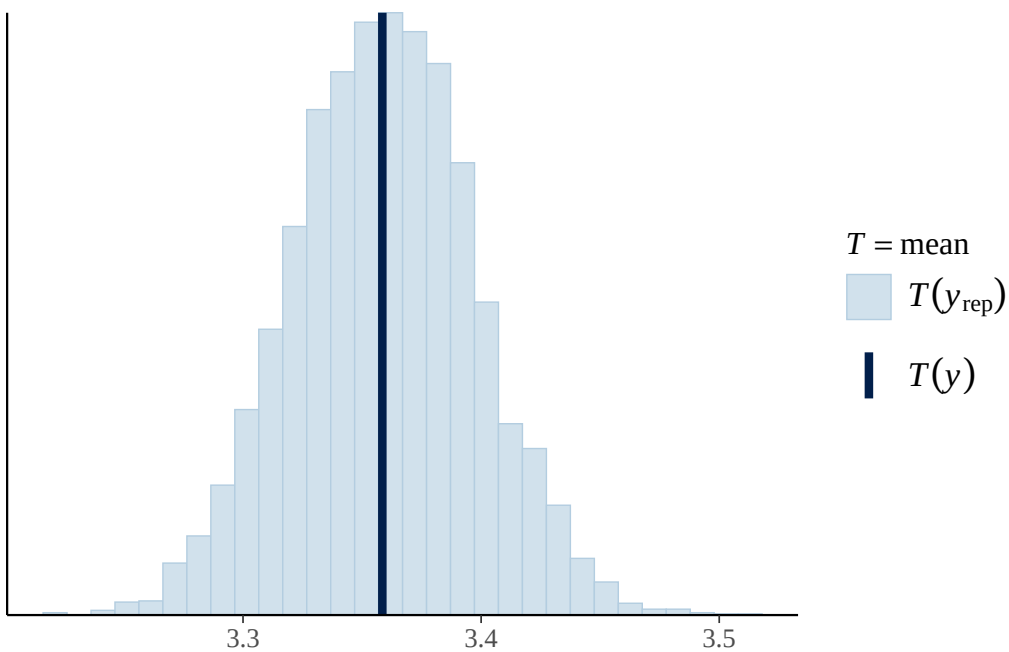
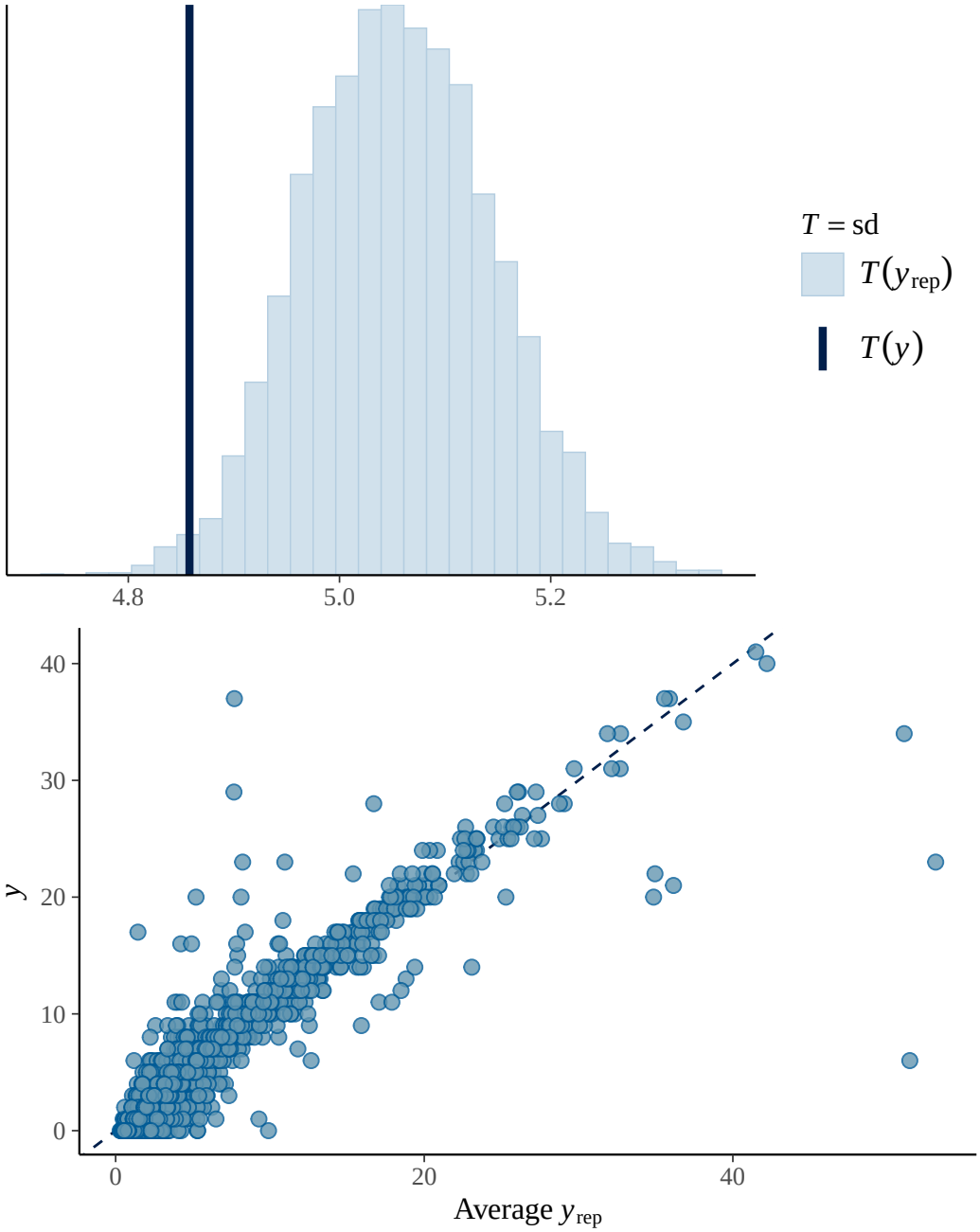

```
`stat_bin()` using `bins = 30`. Pick better value with `binwidth`.
```



```
`stat_bin()` using `bins = 30`. Pick better value with `binwidth`.
```




```
`stat_bin()` using `bins = 30`. Pick better value with `binwidth`.
```



```
`stat_bin()` using `bins = 30`. Pick better value with `binwidth`.
```